

**Comparing Mutual Information in Finite Samples:  
Wald and Expanded Welch–Satterthwaite  
Inference**

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# Abstract

Mutual information (MI) measures association between two variables. This thesis asks whether two independently sampled populations have equal MI, even when their joint probability tables differ. The usual Normal Wald test uses a normal approximation that can be inaccurate in small or sparse samples.

The thesis develops an Expanded Welch–Satterthwaite adjustment for this comparison. It derives how changing one observation affects the estimated MI variance, including changes to the cell probabilities and their row and column totals. That calculation determines the degrees of freedom for a Student  $t$  reference. Expanded Welch retains Wald’s estimated MI difference and standard error; it changes only the distribution used to judge the resulting statistic.

The methods are evaluated in 3,111 fixed statistical settings, each with 20,000 independently sampled table pairs. The design includes square tables from  $2 \times 2$  to  $8 \times 8$ , selected rectangular tables, unequal sample sizes, sparse cells, and populations with different distributions but equal MI. At a nominal 5% threshold, the experiment measures false positives, detection when MI differs, and the frequency of usable results.

Expanded Welch reduces excessive Wald false positives in some settings. However, when both tests return a p-value, its Student reference gives a p-value no smaller than Wald’s. It therefore also lowers rejection where Wald already rejects too rarely and can reduce detection when MI differs. In sparse samples, Expanded Welch more often fails to return a usable p-value under the implemented rule. Wald is closer to the target 5% rate in all 24 main null settings with sample sizes 500 or 1,000. A supplementary diagnostic shows that changing the reference distribution cannot correct error in the standard error shared by both tests. For the settings studied, Wald remains the simpler general baseline; Expanded Welch is an additional check where Wald is known to reject equal MI too often.

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# Chapter 1

## Introduction

Suppose two independent groups are observed on the same categorical variables  $X$  and  $Y$ . Mutual information (MI) describes the strength of association between those variables in each group. If the estimated MI differs, the difference alone does not tell us whether the population MI values differ: both estimates vary when new observations are drawn. The question in this thesis is whether the two *population* MI values can plausibly be equal.

This is not the usual test of whether  $X$  and  $Y$  are independent within one group. The null hypothesis here permits dependence in both groups and permits their complete joint distributions to differ. It specifies only

$$H_0 : I(P) = I(Q), \quad H_1 : I(P) \neq I(Q), \quad (1.1)$$

where  $P$  and  $Q$  are the two joint distributions. The samples from  $P$  and  $Q$  are independent of each other; that sampling assumption does not imply independence of  $X$  and  $Y$  within either population.

### 1.1 Why the comparison is difficult

MI uses the probabilities in every cell of a table and the totals of its rows and columns. In a small sample, some cells can be empty even when their population probabilities are positive. The observed tables determine both the estimated MI difference and its standard error, so sparse data can affect both. The Normal Wald test compares their ratio with a normal curve, but in small samples it may reject equal MI more or less often than the intended 5%.

Expanded Welch offers a possible adjustment. It uses the same estimated difference and standard error as Wald, but compares their ratio with a Student  $t$

distribution. Its degrees of freedom depend on how much the estimated MI variance changes from sample to sample. To calculate that change, we must account for the fact that changing one cell also changes its row and column totals and the MI values calculated from them. This thesis derives that calculation and tests whether the adjustment helps.

The adjustment is not automatically better. It can reduce false positives where Wald rejects too often, but it can also reduce detection where Wald already rejects too rarely. In sparse samples, it may fail to return a usable p-value when Wald succeeds. The experiment therefore examines false positives, detection and the frequency of usable results together.

## 1.2 Research questions

The thesis asks four related questions. First, how can changes in the estimated MI variance be used to choose the degrees of freedom for a Student test? Second, does this choice improve false-positive rates without losing too much detection or producing too many invalid results? Third, how does the answer depend on table size, uneven cell probabilities, the amount and location of association, and unequal sample sizes? Fourth, do the problems disappear as samples grow, and how much extra computation does the method require?

## 1.3 Contributions to the field

This thesis combines established ideas about MI and Welch–Satterthwaite tests for a specific question: whether two populations have equal MI. Its contributions are the calculation needed for that test and evidence about whether the test is useful.

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance. The calculation includes the resulting changes in the row and column totals and in pointwise MI.
2. **Methodological contribution.** Expanded Welch uses this calculation to choose the degrees of freedom for a two-sample Student test. The main experiment compares it with Wald; a supplementary analysis checks simpler sample-size choices for the degrees of freedom.
3. **Empirical contribution.** A reproducible experiment measures false positives, detection, usable results and runtime on fixed populations with known MI. It covers  $2 \times 2$  through  $8 \times 8$  square tables, rectangular tables, unequal samples

and very sparse settings. The companion atlas retains the result for each setting rather than hiding it in an overall average.

4. **Practical contribution.** Expanded Welch can help when Wald rejects equal MI too often. It can also make an already-low rejection rate lower and return fewer usable results in sparse samples. The evidence shows why it should not replace Wald in general and points to problems that changing the reference distribution alone cannot solve.

## 1.4 Scope and organisation

The analysis concerns independent samples recorded in count tables with the same row and column categories. The main derivation assumes every population cell has positive probability and that its leading large-sample MI variance is greater than zero. Exact independence is considered separately because that variance becomes zero. The experiment compares Normal Wald with Expanded Welch using a two-sided 5% test. It does not establish performance for paired data, continuous MI estimators or a number of categories that grows with sample size.

Chapter 2 supplies the statistical background. Chapters 3 and 4 define the shared statistic and derive the correction. Chapter 5 explains how the populations and samples are generated and how the results are measured. Chapter 6 shows the complete landscape before the focused comparisons. Chapters 7 and 8 discuss the consequences and the resulting recommendation. The appendices provide the detailed algebra, construction and source-to-result record.

# Chapter 2

## Background and Literature Review

This chapter develops the concepts needed to compare MI between two categorical populations. It begins with entropy, pointwise MI, and population MI. It then reviews plug-in estimation, first-order inference, the Welch–Satterthwaite approximation, and prior comparisons of information quantities. The final section identifies the gap addressed by the expanded method.

### 2.1 Information-theoretic foundations

#### 2.1.1 Entropy and information content

For a discrete random variable  $X$  with probability mass function  $p_X(x)$ , the information content of outcome  $x$  is  $-\log p_X(x)$ . Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = - \sum_x p_X(x) \log p_X(x). \quad (2.1)$$

Entropy summarises the uncertainty of one random variable before its value is observed ([Shannon, 1948](#)).

For a pair  $(X, Y)$ , the joint entropy  $H(X, Y)$  is defined from the joint probabilities in the same way. The conditional entropy  $H(X | Y) = H(X, Y) - H(Y)$  measures the uncertainty remaining in  $X$  after observing  $Y$ .

### 2.1.2 Pointwise and average mutual information

Let  $P = (p_{ij})$  be the joint distribution of categorical variables  $X \in \{1, \dots, r\}$  and  $Y \in \{1, \dots, c\}$ . Write

$$p_{i+} = \sum_{j=1}^c p_{ij}, \quad p_{+j} = \sum_{i=1}^r p_{ij} \quad (2.2)$$

for the row and column margins. The pointwise mutual information associated with cell  $(i, j)$  is

$$\ell_P(i, j) = \log \left( \frac{p_{ij}}{p_{i+}p_{+j}} \right). \quad (2.3)$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts; a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^r \sum_{j=1}^c p_{ij} \ell_P(i, j). \quad (2.4)$$

Here the summations are nested: for each row  $i$ , the inner sum runs over all columns  $j$ . Equivalently, the expression sums once over every cell in the table and may be written with the shorthand  $\sum_{i,j}$ . Because  $P$  is the joint distribution of  $(X, Y)$ , the same quantity is conventionally written as  $I(X; Y)$ . Equivalently,

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (2.5)$$

The quantity is nonnegative and equals zero exactly when  $p_{ij} = p_{i+}p_{+j}$  for every cell, including zero-probability cells.

MI measures the overall departure from independence, not one particular shape of association.

This scalar summary deliberately discards much of the information in the joint distribution. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. Our null requires only this one number to agree, not that the two tables match cell by cell.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by  $\log 2$  converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

## 2.2 Estimating discrete mutual information

Suppose a sample of size  $n$  produces counts  $N_{ij}$ . The empirical cell probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}}{n}. \quad (2.6)$$

Substitution into Equation (2.4) gives the plug-in estimator

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \log \left( \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right). \quad (2.7)$$

The zero-cell convention follows from  $\lim_{x \downarrow 0} x \log x = 0$ . Empty cells contribute no direct term, but they still affect the empirical margins and the finite-sample behaviour of the estimator.

Under multinomial sampling, the cell counts are random but constrained to sum to  $n$ , so they are dependent. The plug-in estimator remains consistent for a fixed alphabet, but its finite-sample behaviour depends on the complete probability table. In particular, rare cells can change the sampling behaviour even when the alphabet and sample sizes are unchanged (Paninski, 2003).

The plug-in estimator is upward biased: random variation in the cell counts tends to make the empirical table appear more dependent than the population table. For a fixed positive  $r \times c$  support, the leading bias is

$$\text{Bias}\{\hat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \quad (2.8)$$

The factor  $(r-1)(c-1)$  increases with table size, while division by  $n$  shows that the bias decreases as the sample grows. The bias is therefore largest for larger tables with small samples. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing two populations. If  $I(P) = I(Q)$  but  $n_P \neq n_Q$ , the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2} \left( \frac{1}{n_P} - \frac{1}{n_Q} \right). \quad (2.9)$$

For a  $2 \times 2$  table with  $n_P = 50$  and  $n_Q = 100$ , Equation (2.9) gives 0.005 nats despite equal population MIs. Chapter 3 corrects this sample-size effect by subtracting  $(r-1)(c-1)/(2n)$  from each estimate.

## 2.3 Sampling variance of plug-in MI

To compare two estimated MI values, it is necessary to know how much each estimate would vary across repeated samples. The goal of this section is to derive the large-sample approximation

$$\text{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n}. \quad (2.10)$$

Here  $V(P)$ , defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004). The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

### Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (2.11)$$

Let  $f(u) = u \log u$ . The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du} \{u \log u\} = \log u + u \frac{1}{u} = \log u + 1. \quad (2.12)$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \quad (2.13)$$

where  $a$  is the value around which the function is expanded and  $x - a$  is the distance from that value. Here the new value is the estimated probability  $x = \hat{u}$ , the expansion point is the population probability  $a = u$ , and  $f(t) = t \log t$ . Therefore,  $x - a = \hat{u} - u$ , which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\hat{u} \log \hat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\hat{u} - u)}_{\text{first-order change}}. \quad (2.14)$$

Subtracting the population value from both sides makes the change explicit:

$$\hat{u} \log \hat{u} - u \log u \approx (\log u + 1)(\hat{u} - u). \quad (2.15)$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \sum_{i,j} (\log p_{ij} + 1)(\hat{p}_{ij} - p_{ij}) \\ &\quad - \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) \\ &\quad - \sum_j (\log p_{+j} + 1)(\hat{p}_{+j} - p_{+j}). \end{aligned} \quad (2.16)$$

The margin errors are sums of the corresponding cell errors:

$$\hat{p}_{i+} - p_{i+} = \sum_j (\hat{p}_{ij} - p_{ij}), \quad \hat{p}_{+j} - p_{+j} = \sum_i (\hat{p}_{ij} - p_{ij}). \quad (2.17)$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) &= \sum_i \left[ (\log p_{i+} + 1) \left\{ \sum_j (\hat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\hat{p}_{ij} - p_{ij}). \end{aligned} \quad (2.18)$$

The inner sum over  $j$  is the probability error for a fixed row  $i$ . It is multiplied by that row's coefficient  $\log p_{i+} + 1$ ; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way. Consequently, the same cell error  $\hat{p}_{ij} - p_{ij}$  appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned} &(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ &= \log \left( \frac{p_{ij}}{p_{i+} p_{+j}} \right) - 1 \\ &= \ell_P(i, j) - 1. \end{aligned} \quad (2.19)$$

Therefore,

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i, j) - 1\}(\hat{p}_{ij} - p_{ij}). \quad (2.20)$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\hat{p}_{ij} - p_{ij}) = \sum_{i,j} \hat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \quad (2.21)$$

The constant  $-1$  in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}
 \hat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\hat{p}_{ij} - p_{ij}) \\
 &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}) - \sum_{i,j} (\hat{p}_{ij} - p_{ij}) \\
 &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}). \tag{2.22}
 \end{aligned}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

**Step 2: From cell errors to observation-level errors.**

Let  $Z_a = (X_a, Y_a)$  be observation  $a$ . An empirical cell probability is the proportion of observations in that cell, so

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \tag{2.23}$$

Substituting this expression into Equation (2.22) first gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[ \frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\} \right]. \tag{2.24}$$

The finite sums can be exchanged, so the contribution from observation  $a$  is gathered inside one pair of brackets:

$$\begin{aligned}
 \hat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^n \left[ \sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\
 &= \frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}. \tag{2.25}
 \end{aligned}$$

The first sum inside the brackets selects the single cell containing  $Z_a$ , giving  $\ell_P(Z_a)$ . The second is the probability-weighted average of all pointwise-MI values, which is  $I(P)$ . This equation is the key step. To first order, the error in the plug-in MI estimate behaves like the average of  $n$  observation-level contributions. The contribution of an observation in cell  $(i,j)$  is simply that cell's pointwise MI minus the population average,  $\ell_P(i,j) - I(P)$ . Its mean is zero because

$$E_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j} p_{ij} \{\ell_P(i,j) - I(P)\}$$

$$= \underbrace{\sum_{i,j} p_{ij} \ell_P(i, j)}_{I(P)} - I(P) \underbrace{\sum_{i,j} p_{ij}}_1 = 0. \quad (2.26)$$

In formal asymptotic terminology, the centred function  $\ell_P(i, j) - I(P)$  is the influence function of MI; Equation (2.25) is its first-order linear representation (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

**Step 3: Calculate the sampling variance.**

Define the population variance of these centred contributions by

$$V(P) = \text{Var}_P\{\ell_P(X, Y) - I(P)\} = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (2.27)$$

Under independent sampling, the  $n$  terms in Equation (2.25) are independent and each has variance  $V(P)$ . The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned} \text{Var}\{\hat{I}(P)\} &\approx \text{Var}\left\{\frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2} \sum_{a=1}^n \text{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2} \sum_{a=1}^n V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \quad (2.28)$$

This proves the approximation stated in Equation (2.10). It is the familiar result that the variance of an average of  $n$  independent contributions is the variance of one contribution divided by  $n$ .

When  $V(P) > 0$ , the central limit theorem also gives

$$\frac{\hat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0, 1). \quad (2.29)$$

This does not claim that every finite-sample MI estimate is normally distributed. It is a local, large-sample approximation for a fixed table with positive population cell probabilities. The  $1/n$  leading bias from Section 2.2 decreases faster than the  $1/\sqrt{n}$  sampling error, so it does not change this limiting distribution.

**The independence boundary.**

At exact independence,  $p_{ij} = p_{i+}p_{+j}$ , so  $\ell_P(i, j) = \log 1 = 0$  in every cell. Hence  $I(P) = 0$ ,  $V(P) = 0$ , and the first-order average in Equation (2.25) vanishes.

A finite sample can still have positive MI. Its leading variation comes from the

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squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limit ([Brillinger, 2004](#); [Marinescu and Balcau, 2025](#)). The normal limit for comparing positive MI values and the chi-squared limit for testing independence therefore apply in different settings.

## 2.4 Related inferential methods

### Normal Wald inference.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with its asymptotic normal reference (Wald, 1943; van der Vaart, 1998). For two independent MI estimates, the two first-order variance contributions add:

$$\text{Var}\{\hat{I}(P) - \hat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \quad (2.30)$$

Each term is the sampling variance of one MI estimate. Independence of the two samples makes their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\hat{I}(P) - \hat{I}(Q)}{\sqrt{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q}} \xrightarrow{d} N(0, 1) \quad \text{under } I(P) = I(Q). \quad (2.31)$$

Here  $\hat{V}(P)$  and  $\hat{V}(Q)$  estimate the pointwise-MI variances  $V(P)$  and  $V(Q)$ . This limit assumes the regular positive-support conditions of Section 2.3 and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test.

With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses a normal reference; it makes no separate small-sample adjustment for that standard-error variability.

### The Welch–Satterthwaite principle.

Welch’s test compares two means when their population variances may differ (Welch, 1947). For two independent samples with means  $\bar{X}_P, \bar{X}_Q$  and variances  $s_P^2, s_Q^2$ , the statistic for testing equal population means and its effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \quad (2.32)$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\frac{(s_P^2/n_P)^2}{n_P - 1} + \frac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \quad (2.33)$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are  $n_P - 1$  and  $n_Q - 1$ . If  $s_P^2/n_P$  dominates, the formula makes the combined  $\nu$  close to  $n_P - 1$ ; when both contributions matter, both component values affect  $\nu$ . Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

Welch compares the statistic with a Student distribution using the calculated  $\nu$ . At the same significance level, this requires a larger absolute statistic to reject the null than a standard normal reference does. The difference between the Student and normal rejection thresholds becomes smaller as  $\nu$  increases.

For MI, the estimated variance contributions are  $\hat{V}(P)/n_P$  and  $\hat{V}(Q)/n_Q$ . Their component degrees of freedom need their own calculation because  $\hat{V}(P)$  and  $\hat{V}(Q)$  are nonlinear functions of the probability tables. Chapter 4 derives these MI-specific inputs to the same moment-matching structure.

Regression studies have also adjusted reference distributions to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). They establish the general strategy, but not whether it helps with MI. For example, Bell and McCaffrey (2002) report regression variance estimates biased downward, while Kauermann and Carroll (2001) examine intervals that cover too rarely. A more cautious reference can help in such settings. Near independence, however, the estimated MI variance can be biased upward. A Student reference may then make an already cautious test reject even less often. Section 6.7 examines this possibility.

### Prior MI inference and related comparisons.

MI uncertainty also has a broader literature than its leading normal approximation. Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. Hutter (2002) and Hutter and Zaffalon (2005) derive moments of the posterior distribution of MI under a Dirichlet model, including higher-order variance and shape information. Their posterior uncertainty has a different conditioning from the repeated-sampling distribution studied here. Their results establish that higher-order MI uncertainty is not a new topic, but do not directly validate the two-sample Student reference in this thesis.

Hutcheson's test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student reference, using

$$\nu_H = \frac{(s_{H,P}^2 + s_{H,Q}^2)^2}{(s_{H,P}^2)^2/n_P + (s_{H,Q}^2)^2/n_Q} \quad (2.34)$$

as its effective degrees of freedom (Hutcheson, 1970). Here each of  $s_{H,P}^2$  and  $s_{H,Q}^2$  is the estimated sampling variance of the corresponding entropy estimator  $\widehat{H}$ , and each variance component is assigned its sample size, not  $n_i - 1$ . Applying the same assignment to the two MI variance contributions gives a clear historical baseline. It does not, however, derive how the plug-in MI variance itself fluctuates: MI depends on a joint distribution and both margins, so moving mass into one cell changes pointwise values elsewhere in its row and column. The expanded calculation retains these changes.

The question of comparing MI values also has predecessors. In a segregation application, Mora and Ruiz-Castillo (2009) develop inference for an MI index and discuss pairwise comparisons across groups and periods; the later peer-reviewed account places that MI index within the wider segregation literature (Mora and Ruiz-Castillo, 2011). They do not derive the variance-sensitivity degrees of freedom used here, but the general aim of comparing MI values is not new. Guha and Chothia (2014) use MI for a different two-sample question: whether two continuous distributions are identical. Their MI is between a pooled observation and its sample label; their null is equality of distributions, not equality of two within-population MI values. For discrete data, Drake and Guha (2014) develop an MI-based  $k$ -sample test for the same distributional-equality question.

At exact independence, the first-order sampling variation of MI vanishes and its second-order variation becomes important (Marinescu and Balcau, 2025). The positive-MI comparison developed here is therefore not a replacement for a one-table independence test. Second-order effects can also matter near that boundary.

### Resampling alternatives.

A bootstrap would draw many new samples from the observed tables to estimate how much the test statistic varies. This requires repeating the calculation many times (Efron, 1979). A simple permutation test instead combines the two samples and shuffles their group labels. That shuffle is justified when the populations have the same complete distribution,  $P = Q$ . Our hypothesis requires only  $I(P) = I(Q)$ : the tables can differ while their MI values are equal. Shuffling labels could therefore test a stronger claim than this thesis asks about (Chung and Romano, 2013). More specialised permutation methods are possible for some such comparisons (Janssen, 1997; Chung and Romano, 2013), but assessing one for equal MI would be a separate study. The present comparison focuses on Normal Wald and Expanded Welch.

## 2.5 Previous work and this thesis's contribution

Previous research explains how to estimate MI, correct its leading small-sample bias and approximate its sampling variance. Wald's method uses that variance to compare two estimates. Welch–Satterthwaite methods also allow for uncertainty in an estimated variance. Researchers have also compared entropy and MI-related quantities.

The missing step for this test is to measure how much the *estimated MI variance* changes from sample to sample. Assigning each population degrees of freedom based only on its sample size, as earlier information-based and Welch tests do, does not measure this change. A new observation affects a cell probability, its row and column totals, and the MI values calculated from them. This thesis derives how those changes affect the estimated variance and uses the result to choose a Student  $t$  distribution for comparing two MI values. It then tests the method on fixed population tables, showing when it brings false-positive rates closer to the target and when it instead rejects too rarely.

Chapter 3 defines the comparison, Chapter 4 derives the adjustment, and Chapter 6 evaluates it.

## Chapter 3

# The Two-Sample MI Test and Wald Baseline

This chapter defines the population MI difference being tested and the statistic calculated from the observed tables. Both methods use this same statistic. Normal Wald judges it against a standard normal distribution; Chapter 4 develops a Student distribution with estimated degrees of freedom.

### 3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \quad N^Q = (N_{ij}^Q) \quad (3.1)$$

be independent  $r \times c$  count tables with totals  $n_P$  and  $n_Q$ . The row and column categories have the same definitions in both populations. The empirical probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \quad (3.2)$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \quad (3.3)$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \quad (3.4)$$

The large-sample argument uses the following assumptions.

**Assumption 1** (Sampling). *Observations are independent and identically dis-*

tributed within each population, and the two population samples are independent.

**Assumption 2** (Aligned fixed alphabets). *The row and column categories are fixed in advance and aligned between the two tables.*

**Assumption 3** (Positive cell probabilities). *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

**Assumption 4** (Positive MI). *Both populations have positive MI:  $I(P) > 0$  and  $I(Q) > 0$ . With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances  $V(P) > 0$  and  $V(Q) > 0$ . It excludes exact independence, where the first-order approximation breaks down.*

**Assumption 5** (Logarithm units). *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

### Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\hat{\ell}_P(i, j) = \log \left( \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right) \quad \text{when } \hat{p}_{ij} > 0. \quad (3.5)$$

The plug-in MI estimate is

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \hat{\ell}_P(i, j). \quad (3.6)$$

The same calculation gives  $\hat{I}(Q)$ .

The leading fixed-support MI bias appears in Equation (2.8) (Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\hat{\Delta}_{\text{BC}} = \left\{ \hat{I}(P) - \frac{(r-1)(c-1)}{2n_P} \right\} - \left\{ \hat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q} \right\}. \quad (3.7)$$

The two terms in braces are the bias-corrected MI estimates for the separate populations.

The correction uses the declared alphabet dimensions, not the observed nonempty dimensions. A corrected estimate can be negative in a sparse sample and is not clipped at zero, since clipping would define a different nonlinear estimator. When

$n_P = n_Q$ , the two corrections cancel in the difference; otherwise, their difference removes the leading unequal-sample bias described in Section 2.2.

## 3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of  $\hat{I}(P)$  is  $V(P)/n_P$ , where  $V(P)$  is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates  $V(P)$  by

$$\hat{V}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \{\hat{\ell}_P(i,j) - \hat{I}(P)\}^2. \quad (3.8)$$

Define  $\hat{V}(Q)$  analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\text{SE}}^2 = \frac{\hat{V}(P)}{n_P} + \frac{\hat{V}(Q)}{n_Q}. \quad (3.9)$$

The common standardised statistic is

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\widehat{\text{SE}}}. \quad (3.10)$$

The null hypothesis is not imposed when calculating  $\hat{\Delta}_{\text{BC}}$ ,  $\widehat{\text{SE}}$  or  $T$ : they use the observed tables. Equal population MI enters when interpreting the reference distribution, which is centred at zero under  $H_0$ . Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

## 3.3 Normal Wald reference and motivation for expansion

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0, 1) \quad \text{under } H_0. \quad (3.11)$$

The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \quad (3.12)$$

where  $\Phi$  is the standard normal distribution function.

The normal reference is a large-sample approximation. Wald estimates the standard error from the tables but applies no degrees-of-freedom adjustment.

### Why consider an expanded reference?

The shared statistic supplies the estimated effect and its first-order standard error. Earlier information-theoretic and ordinary-Welch analogues set each variance component's degrees of freedom from its sample size:  $n_i$  (Hutcheson, 1970) or  $n_i - 1$  (Welch, 1947). Neither assignment measures how much the plug-in MI variance changes when the observed table changes. That variance depends on joint cells, margins and pointwise MI values. Expanded Welch instead estimates each component's degrees of freedom from how the complete MI variance estimate changes with the table. The supplementary analysis also compares these sample-size assignments with Expanded Welch.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For  $P$ , to first order,

$$E\{\hat{V}(P)\} \approx V(P). \quad (3.13)$$

Chapter 4 derives  $\text{Var}\{\hat{V}(P)\}$  by tracking how the calculated MI variance changes when one observation changes the joint table.

# Chapter 4

## The Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3. It changes the distribution used to judge that statistic from normal to Student. To choose the Student degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples.

### 4.1 Choosing Degrees of Freedom for Expanded Welch

For the two MI estimates, the Welch–Satterthwaite equation combines their estimated variance contributions as in the original moment-matching framework (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\left\{ \hat{V}(P)/n_P \right\}^2 / \nu_P + \left\{ \hat{V}(Q)/n_Q \right\}^2 / \nu_Q}. \quad (4.1)$$

The numerator contains the two terms in the squared standard error. The remaining quantities are the component degrees of freedom  $\nu_P$  and  $\nu_Q$ , which should measure how reliably  $\hat{V}(P)$  and  $\hat{V}(Q)$  estimate their population counterparts.

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of  $\hat{V}(P)$ . Second, we calculate how  $V(P)$  changes when a small amount of probability moves into one cell. Third, that change tells us how variable  $\hat{V}(P)$  is. Finally, we combine the degrees of freedom from  $P$  and  $Q$  using Equation (4.1).

**Moment matching for one component.**

Satterthwaite represents a positive variance estimate by a scaled chi-squared variable. For the first population, the working approximation represents  $\widehat{V}(P)$  by

$$\mathbb{E}\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}. \quad (4.2)$$

The approximation matches moments; it does not assert that the finite-sample distribution of  $\widehat{V}(P)$  is exactly chi-squared.

For a chi-squared variable with  $\nu_P$  degrees of freedom,

$$\mathbb{E}(\chi_{\nu_P}^2) = \nu_P, \quad \text{Var}(\chi_{\nu_P}^2) = 2\nu_P. \quad (4.3)$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathbb{E} \left[ \mathbb{E}\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P} \right] = \mathbb{E}\{\widehat{V}(P)\}. \quad (4.4)$$

Its variance is

$$\text{Var} \left[ \mathbb{E}\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P} \right] = \frac{[\mathbb{E}\{\widehat{V}(P)\}]^2}{\nu_P^2} \text{Var}(\chi_{\nu_P}^2) \quad (4.5)$$

$$= \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\nu_P}. \quad (4.6)$$

Matching this quantity to  $\text{Var}\{\widehat{V}(P)\}$  gives

$$\boxed{\nu_P = \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\text{Var}\{\widehat{V}(P)\}}}. \quad (4.7)$$

To first order,  $\mathbb{E}\{\widehat{V}(P)\} \approx V(P)$ . The required calculation is therefore the sampling variance of  $\widehat{V}(P)$ .

**4.2 How the MI Variance Changes with the Table**

To find how much the estimated MI variance changes between samples, we first ask how the population variance responds when a small amount of probability is moved into one cell (Hampel, 1974; van der Vaart, 1998). Fix a cell  $(x, y)$  and define the perturbed distribution

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(x,y)}, \quad (4.8)$$

where  $\delta_{(x,y)}$  places probability one on cell  $(x, y)$ . At  $\varepsilon = 0$ , the distribution is  $P$ . Increasing  $\varepsilon$  moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell  $(i, j)$ ,

$$p_{ij}(\varepsilon) = (1 - \varepsilon)p_{ij} + \varepsilon \mathbf{1}\{i = x, j = y\}. \quad (4.9)$$

The joint and marginal probability derivatives at the original distribution are

$$\left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \quad (4.10)$$

$$\left. \frac{dp_{i+}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \quad (4.11)$$

$$\left. \frac{dp_{+j}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \quad (4.12)$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \quad (4.13)$$

### Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \quad (4.14)$$

Differentiating the three logarithms and substituting the probability derivatives gives

$$\left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j) \right|_0 = \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1. \quad (4.15)$$

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j). \quad (4.16)$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned} \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \{ \mathbf{1}(i=x, j=y) - p_{ij} \} \ell_P(i, j) \\ &\quad + \sum_{i,j} p_{ij} \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.17)$$

The first sum is  $\ell_P(x, y) - I(P)$ . In the second sum, the selected-cell term contributes  $p_{xy}/p_{xy} = 1$ . The row term contributes  $-\sum_j p_{xj}/p_{x+} = -1$ ; the column term likewise contributes  $-1$ ; and the constant contributes  $\sum_{i,j} p_{ij} = 1$ . These four contributions cancel. Hence

$$\boxed{\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P)}. \quad (4.18)$$

This result recovers the centred pointwise-MI influence function used in Chapter 3.

#### Effect on the MI variance.

The target is the first-order change in

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\}. \quad (4.19)$$

Write this variance in second-moment form:

$$V(P) = \text{E}_P\{\ell_P(X, Y)^2\} - I(P)^2. \quad (4.20)$$

Define the observation-level variance sensitivity by

$$g_P(x, y) = \left. \frac{dV(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.21)$$

Differentiating Equation (4.20) gives

$$g_P(x, y) = \left. \frac{d}{d\varepsilon} \text{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} \right|_0 - 2I(P) \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.22)$$

It remains to differentiate

$$\text{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j)^2. \quad (4.23)$$

The product rule gives

$$\begin{aligned} \left. \frac{d}{d\varepsilon} E_{P_\varepsilon} \{ \ell_{P_\varepsilon}(X, Y)^2 \} \right|_0 &= \sum_{i,j} \left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 \ell_P(i, j)^2 \\ &\quad + 2 \sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.24)$$

The first sum is

$$\ell_P(x, y)^2 - E_P \{ \ell_P(X, Y)^2 \}. \quad (4.25)$$

For the second sum, define the probability-weighted row and column means

$$E_P \{ \ell_P(X, Y) \mid X = x \} = \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}}, \quad (4.26)$$

$$E_P \{ \ell_P(X, Y) \mid Y = y \} = \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}}. \quad (4.27)$$

Substitution of Equation (4.15) gives

$$\begin{aligned} &\sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0 \\ &= \ell_P(x, y) - \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}} - \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}} + I(P) \\ &= \ell_P(x, y) - E_P \{ \ell_P(X, Y) \mid X = x \} - E_P \{ \ell_P(X, Y) \mid Y = y \} + I(P). \end{aligned} \quad (4.28)$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) yields

$$\boxed{\begin{aligned} g_P(x, y) &= \{ \ell_P(x, y) - I(P) \}^2 - V(P) \\ &\quad + 2 \left[ \ell_P(x, y) - E_P \{ \ell_P(X, Y) \mid X = x \} \right. \\ &\quad \left. - E_P \{ \ell_P(X, Y) \mid Y = y \} + I(P) \right]. \end{aligned}} \quad (4.29)$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for changes in those values, including the effects of the selected joint probability and both margins.

## 4.3 Component degrees of freedom and final test

**Population component degrees of freedom.**

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \text{Var}_P\{g_P(X, Y)\}. \quad (4.30)$$

Averaging the one-cell perturbation directions gives  $\sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\} = 0$ . The derivative is linear in its direction, so  $E_P\{g_P(X, Y)\} = 0$ ; Appendix A shows the steps. Explicit centring protects the numerical calculation if roundoff produces a small nonzero empirical mean.

**Assumption 6** (Regular variance sensitivity). *For the first-order Expanded Welch construction,  $\tau^2(P) > 0$  and  $\tau^2(Q) > 0$ .*

The empirical distribution is

$$\hat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\delta_{Z_a^{(P)}} - P\}. \quad (4.31)$$

Taylor's first-order term is linear in this difference (van der Vaart, 1998, Chapter 20). By Equation (4.21), each summand has sensitivity  $g_P(Z_a^{(P)})$ . Therefore,

$$\hat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \quad (4.32)$$

The observations are independent, so

$$\text{Var}\{\hat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \text{Var}_P\{g_P(Z_a^{(P)})\} \quad (4.33)$$

$$= \frac{\tau^2(P)}{n_P}. \quad (4.34)$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \quad (4.35)$$

Equation (4.35) depends on relative, not absolute, uncertainty. Since

$$\frac{\text{Var}\{\hat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \quad (4.36)$$

the component degrees of freedom are approximately twice the reciprocal of this relative variance. A stable estimate relative to the size of  $V(P)$  has large degrees of freedom; a relatively unstable estimate has smaller degrees of freedom and produces a heavier-tailed final reference. This first-order component formula assumes  $\tau^2(P) > 0$ .

Positive  $V(P)$  alone does not guarantee that condition. A small change in the table can have no first-order effect on  $V(P)$ , even when MI and  $V(P)$  are positive. In that case, a higher-order analysis of  $\hat{V}(P)$  would be needed to describe its finite-sample distribution.

### Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \hat{g}_P(i, j) = & \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \\ & + 2 \left[ \hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right]. \end{aligned} \quad (4.37)$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \left\{ \hat{g}_P(i, j) - \sum_{k,l} \hat{p}_{kl} \hat{g}_P(k, l) \right\}^2. \quad (4.38)$$

The estimated component degree of freedom is

$$\boxed{\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}}. \quad (4.39)$$

Repeating the same calculation for the second table gives  $\hat{\nu}_Q$ .

### Combining the two components.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\hat{\nu}_{\text{expanded}} = \frac{\{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q\}^2}{\{\hat{V}(P)/n_P\}^2 / \hat{\nu}_P + \{\hat{V}(Q)/n_Q\}^2 / \hat{\nu}_Q}}. \quad (4.40)$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2 \left\{ 1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right\}, \quad (4.41)$$

where  $F_{t_\nu}$  denotes the cumulative distribution function of a Student  $t$  variable with  $\nu$  degrees of freedom.

The expanded calculation does not change  $\hat{\Delta}_{\text{BC}}$ ,  $\widehat{\text{SE}}$ , or  $T$ . It changes only the

reference distribution used to evaluate that statistic.

For a fixed finite  $T$ , the Student reference gives a two-sided p-value at least as large as the Normal Wald reference. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid. Thus every Expanded Welch rejection is also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. The correction cannot raise rejection where Wald is already too conservative. Appendix A proves this ordering and examines a zero-component case in the combined degrees-of-freedom formula.

## 4.4 Worked example and implementation

Consider the two observed  $3 \times 3$  tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \quad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \quad (4.42)$$

Their totals are  $n_P = 105$  and  $n_Q = 100$ . The plug-in MI estimates are 0.13693 and 0.25848 nats. Since  $(3 - 1)(3 - 1) = 4$ , the corrected difference is

$$\hat{\Delta}_{\text{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \quad (4.43)$$

The empirical influence variances are  $\hat{V}(P) = 0.27189$  and  $\hat{V}(Q) = 0.43371$ . Hence

$$\widehat{\text{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \quad T = -1.44898. \quad (4.44)$$

Normal Wald gives  $p = 0.14734$ .

For Expanded Welch, the table-based sensitivity calculation first gives  $\hat{\tau}^2(P) = 0.93212$  and  $\hat{\tau}^2(Q) = 0.82035$ . Substitution into Equation (4.39) gives component degrees of freedom

$$\hat{\nu}_P = 16.65, \quad \hat{\nu}_Q = 45.86. \quad (4.45)$$

Equation (4.40) combines them into  $\hat{\nu}_{\text{expanded}} = 59.03$ , giving  $p = 0.15264$ . Both methods fail to reject at the 5% level for these tables. The example shows that Expanded Welch changes the reference, not the observed statistic.

### Operational algorithm and complexity.

Table 4.1 summarises one comparison.

**Table 4.1:** Operational Expanded Welch–Satterthwaite algorithm.

| Step | Calculation                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------|
| 1    | Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption. |
| 2    | Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.                         |
| 3    | Compute $\hat{V}(P)$ , $\hat{V}(Q)$ , the shared standard error, and $T$ .                                                    |
| 4    | Compute the row and column pointwise-MI means needed in $\hat{g}_P$ and $\hat{g}_Q$ .                                         |
| 5    | Compute $\hat{\tau}^2(P)$ , $\hat{\tau}^2(Q)$ , and the two component degrees of freedom.                                     |
| 6    | Combine the components using Equation (4.40).                                                                                 |
| 7    | Return the Student p-value, effective degrees of freedom, and validity diagnostics.                                           |

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore  $O(rc)$ , and the working memory is also  $O(rc)$ . The method is deterministic once the two count tables are given.

## 4.5 Behaviour and limitations

### Near independence.

The adjustment has a local explanation. Let  $P_t = P_0 + tH$ , where  $P_0 = (q_{ij})$  is independent with positive cells  $q_{ij} = a_i b_j$ , and  $H$  has zero row and column sums. Thus  $t = 0$  is the expansion point and the margins stay fixed. For  $f(t) = I(P_t)$ , the second-order Taylor expansion has  $f(0) = 0$ ,  $f'(0) = \sum_{i,j} H_{ij} = 0$ , and  $f''(0) = \sum_{i,j} H_{ij}^2 / q_{ij}$ .

$$I(P_t) = \underbrace{0}_{\text{population value}} + \underbrace{0t}_{\text{first order}} + \underbrace{\frac{t^2}{2} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}}}_{\text{second order}} + O(t^3), \quad (4.46)$$

$$\ell_{P_t}(i, j) = \log \left( 1 + \frac{tH_{ij}}{q_{ij}} \right) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \quad (4.47)$$

$$V(P_t) = t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (4.48)$$

Appendix D gives the second-order MI result when the margins can also change. The row and column conditional means of pointwise MI have no first-order term because the corresponding sums of  $H$  are zero. Equation (4.29) then gives  $g_{P_t}(i, j) =$

$2tH_{ij}/q_{ij} + O(t^2)$ , and hence

$$\tau^2(P_t) = 4t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (4.49)$$

For a distribution close to independence on this path, these expansions give

$$V(P) \approx 2I(P), \quad \tau^2(P) \approx 4V(P). \quad (4.50)$$

Substitution into the component formula then gives

$$\nu_P = \frac{2n_P V(P)^2}{\tau^2(P)} \approx n_P I(P). \quad (4.51)$$

Thus low MI can produce a small effective degree of freedom and a much heavier-tailed reference. The approximation is local: it assumes fixed table dimension, positive probabilities and sufficiently regular cell counts. Very near independence, quadratic sampling variation can still be important, so a first-order Student adjustment need not improve finite-sample calibration.

### Theoretical status and validity conditions.

Equations (4.18) and (4.29) describe how MI and its estimated variance change when the underlying table changes slightly. For tables with positive cell probabilities, these changes give their first-order large-sample approximations.

The scaled chi-squared model in Equation (4.2) and the Student reference are approximations, not exact finite-sample distributions. In particular,  $\hat{I}(P)$  and  $\hat{V}(P)$  come from the same sample and can vary together. Adjusting the reference for uncertainty in  $\hat{V}(P)$  does not account fully for this dependence or make the statistic exactly Student distributed. Appendix A gives the first-order covariance. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student reference approaches the normal reference used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error. Expanded Welch also needs positive, finite degrees of freedom for each population and for their combination. If a requirement fails, the calculation is recorded as invalid, not as evidence for the null. Appendix E gives the exact software rules. Chapters 5 and 6 report how often valid results are obtained and whether the correction improves calibration.

# Chapter 5

## Experimental Design

The purpose of the experiment is to compare the calibration, detection and validity of two tests across controlled populations. Each design point fixes two population distributions in advance. Repeated independent samples from those populations show how often each procedure returns a valid result and how often it rejects. Because the population MI values are known, a rejection at equal MI is a false positive; a rejection at unequal MI is detection.

### 5.1 A Concrete Pair of Population Tables

Consider two  $2 \times 2$  populations whose row and column margins are  $(0.7, 0.3)$  for  $P$  and  $(0.8, 0.2)$  for  $Q$ . The construction produces, to the displayed precision,

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \quad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \quad (5.1)$$

These are *probability* tables, not observed counts. Their target MI values are 0.02 and 0.03 nats, so the true magnitude of the difference is 0.01 nats. Sampling new count tables changes the estimated MI values, but does not move this fixed population pair to a new horizontal-axis location. Calculations use the stored unrounded probabilities; the rounded display in Equation (5.1) does not determine the samples.

### 5.2 Constructing fixed joint distributions

For one population, let  $a = (a_1, \dots, a_r)$  and  $b = (b_1, \dots, b_c)$  be prescribed row and column probabilities. The independence table has cell probabilities  $a_i b_j$ . Association

is introduced by

$$R(t) = ab^T + tH, \quad H\mathbf{1}_c = 0, \quad \mathbf{1}_r^T H = 0. \quad (5.2)$$

The zero row and column sums mean that changing  $t$  leaves the margins fixed. For the main construction,  $H$  is zero except in the first two rows and columns, where it is

$$H_{1:2,1:2} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (5.3)$$

Thus  $t$  is added to the two diagonal cells and removed from the other two cells. The changes cancel within each affected row and column. The same matrix specifies a  $2 \times 2$  association and a localised change in a larger table; only its embedding differs. This follows the minimal four-violation sparse alternative used by [Berrett and Samworth \(2021, Section 3\(b\)\)](#). Here it is adapted to compare two fixed populations: their margins are prescribed separately and the perturbation strength is chosen to attain a specified MI and is not used directly as the horizontal-axis effect.

For the pair in Equation (5.1), the starting independence tables are

$$P_0 = \begin{pmatrix} 0.49 & 0.21 \\ 0.21 & 0.09 \end{pmatrix}, \quad Q_0 = \begin{pmatrix} 0.64 & 0.16 \\ 0.16 & 0.04 \end{pmatrix}. \quad (5.4)$$

Adding the block in Equation (5.3) with strengths  $t_P \simeq 0.04273$  and  $t_Q \simeq 0.04175$ , respectively, gives the displayed  $P$  and  $Q$ . The strengths are chosen to reach their separate MI targets; the 0.01-nat difference is not a cell-probability increment.

The largest nonnegative strength before a cell with  $H_{ij} < 0$  reaches zero is

$$t_{\max} = \min_{H_{ij} < 0} \frac{a_i b_j}{-H_{ij}}. \quad (5.5)$$

The construction solves for  $0 \leq t < t_{\max}$  such that  $I\{R(t)\}$  equals the requested target. Targets at or beyond the endpoint are not admitted. This endpoint is specific to the chosen path, not the largest possible MI over every table with those margins. Both constructed tables are checked for strictly positive cells, correct margins and target MI before any sampling. Appendix B gives the numerical details.

For a dimension of size  $k$ , the skewed margin used in the design is

$$m_k(\rho) = \left( \rho, \frac{1-\rho}{k-1}, \dots, \frac{1-\rho}{k-1} \right). \quad (5.6)$$

The three principal profiles are specified separately for the two populations in

Table 5.1. For a rectangular  $r \times c$  table, the formula is applied with  $k = r$  to row margins and  $k = c$  to column margins. At a zero MI difference, the same-margin profiles produce  $P = Q$  under this construction; the different-skew profile has  $I(P) = I(Q)$  but  $P \neq Q$ .

**Table 5.1:** Row and column margins used for each population in the principal design. “Uniform” means the vector with every entry equal to  $1/k$ .

| Profile        | $P$ : rows and columns | $Q$ : rows and columns |
|----------------|------------------------|------------------------|
| Uniform        | uniform                | uniform                |
| Same skew      | $m_k(0.8)$             | $m_k(0.8)$             |
| Different skew | $m_k(0.7)$             | $m_k(0.8)$             |

In the usual effect direction,  $I(P) = I_0$  and  $I(Q) = I_0 + |\Delta|$ . The paired solve chooses separate strengths for  $P$  and  $Q$ ; it does not add random noise to their true MI values. The reverse-direction imbalance experiment assigns the higher target to  $P$ . The statistical convention remains  $\Delta = I(P) - I(Q)$ , which is negative in the usual direction. Figures use the nonnegative magnitude  $|\Delta|$ .

Two focused checks ask whether results depend on where the association sits in the table. The “rare block” moves the four changed cells from the upper-left corner to rows and columns with smaller probabilities in skewed populations. The “spread” pattern changes cells throughout the table while keeping the row and column totals fixed. The log-linear comparison also spreads association through the table, using a different rule. In each case, the population is built to have the specified margins and target MI. Tables can therefore agree on margins and MI while assigning different probabilities to individual cells. Appendix B gives the three construction formulas.

## 5.3 Sampling, tests and measured outcomes

For each unique configuration, 20,000 independent pairs of multinomial count tables are drawn from its fixed  $P$  and  $Q$ . Normal Wald and Expanded Welch are applied to the *same* count-table pair. This pairing makes method differences more precisely measurable than unrelated simulations, though it does not remove Monte Carlo uncertainty. The two-sided nominal threshold is  $\alpha = 0.05$  throughout the statistical study.

For each method, the primary plotted rejection rate is

$$\frac{\#\{\text{sample pairs giving a valid p-value no larger than } 0.05\}}{20,000}. \quad (5.7)$$

At  $|\Delta| = 0$  this estimates the false-positive rate. At  $|\Delta| > 0$  it estimates the probability of detecting the specified difference at the fixed nominal threshold, often called power. A rate near 0.05 is desirable under the null; a smaller rate is not automatically better. Under an alternative, higher detection is useful only alongside acceptable null calibration and a usable test result.

An invalid calculation contributes no rejection to Equation (5.7), but it does not count as evidence for the null. The companion valid-result rate shows how often each test returns a usable p-value. The reported 95% intervals show simulation uncertainty at each fixed population pair. Intervals for the difference between methods account for applying both tests to the same sampled tables. Appendix C gives the interval calculations and rates restricted to valid results.

The “minimum expected count” diagnostic is  $\min_{ij} n_{Pp_{ij}}$  for  $P$ , or  $\min_{ij} n_{Qq_{ij}}$  for  $Q$ . It is not the expectation of the smallest observed count. No lower bound on that diagnostic was used to admit statistical configurations. Empty observed cells and rows remain possible. In the plots a hollow marker warns that fewer than 90% of outputs of that method were valid; 90% is a display convention, not a validated safety threshold.

## 5.4 Experiment families and computation

Table 5.2 records the question answered by each family. The common population construction above applies unless the named factor changes. Exact parameter sets are in Appendix B, and the machine-readable protocol is identified in Appendix E. There are 534 distinct population pairs  $(P, Q)$ , each specifying two fixed joint probability tables. Reusing a pair at different sample sizes creates distinct sampling configurations. There are 3,111 unique statistical configurations but 4,001 display points: some configurations are deliberately shown in more than one comparison. Each unique configuration is sampled once, giving 62,220,000 table pairs in the final statistical run. The within-family counts in Table 5.2 do not sum to 3,111 because the same configuration can occur in several families. Display reuse is not an independent replication.

The runtime experiment measures a full method call after count tables are already available. It varies table shape, profile, sample size and null or alternative status. Each regime uses 200 saved inputs, 20 warm-up calls and five timed passes. Thus it measures method computation, not the cost of constructing populations or generating random tables. Median call times and interquartile ranges are reported in Chapter 6.

**Table 5.2:** The fixed experiment families. All statistical points use 20,000 paired replicates and the 5% threshold.

| Family           | Displays | Unique | Factor examined                                                 |
|------------------|----------|--------|-----------------------------------------------------------------|
| Main             | 648      | 648    | Square table size, three margin profiles and equal sample size. |
| Baseline         | 648      | 648    | Starting positive MI of 0.0001, 0.001 or 0.02 nats.             |
| Changed cells    | 144      | 144    | First, rare or spread association patterns.                     |
| Imbalance        | 1,152    | 1,056  | Sample allocation and which population has higher MI.           |
| Larger effects   | 162      | 162    | MI differences up to 0.2 nats where feasible.                   |
| Rectangular      | 108      | 108    | $2 \times 3$ and $3 \times 5$ tables.                           |
| Construction     | 432      | 432    | Additive versus ordinal log-linear population tables.           |
| Extreme skew     | 315      | 315    | Dominant margins up to 0.9995, including $n = 1$ stress.        |
| Rare-cell stress | 140      | 140    | First- versus rare-block changes, including $n = 1$ stress.     |
| Independence     | 144      | 144    | Exact $I(P) = 0$ boundary.                                      |
| Convergence      | 108      | 108    | Fixed equal-MI populations with sample sizes up to 50,000.      |
| Runtime          | 60       | 60     | Full-call timing regimes, separate from simulation.             |

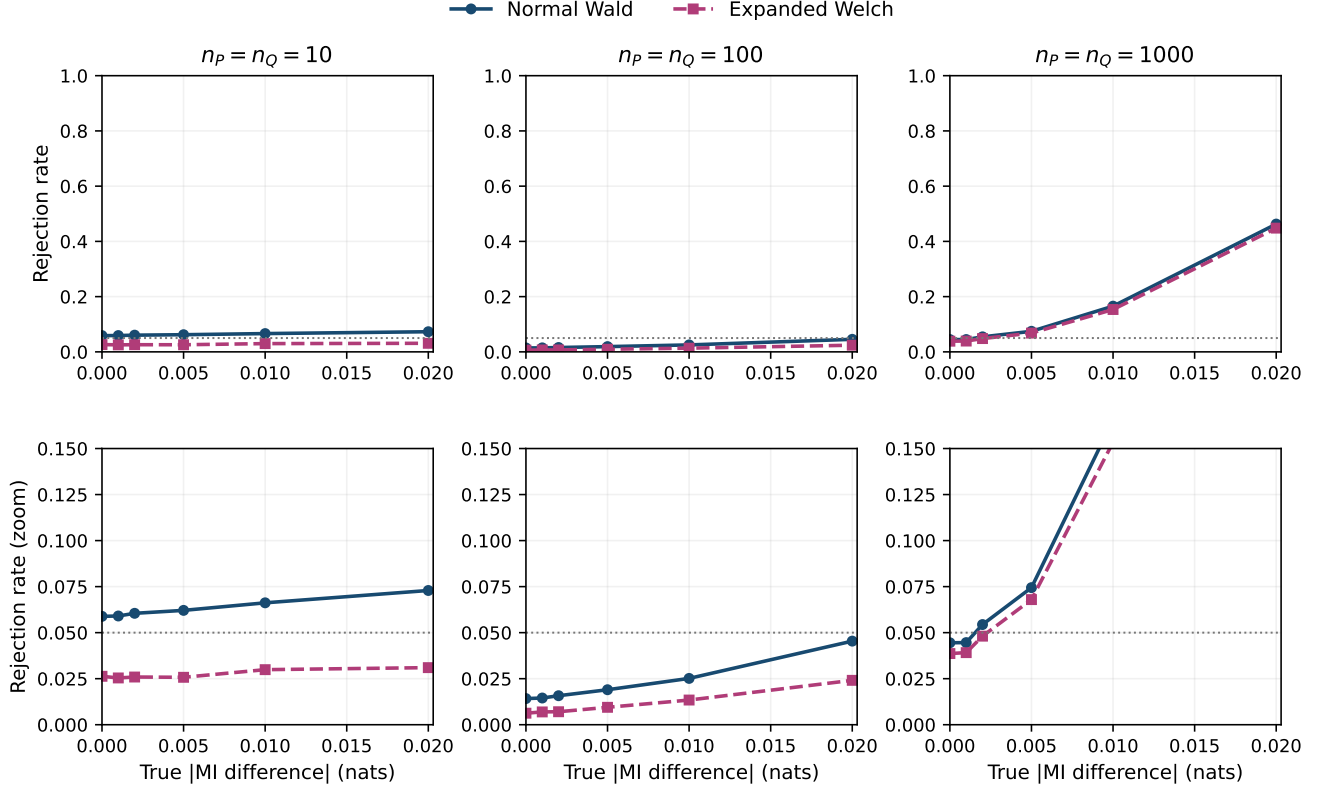
# Chapter 6

## Experimental Results

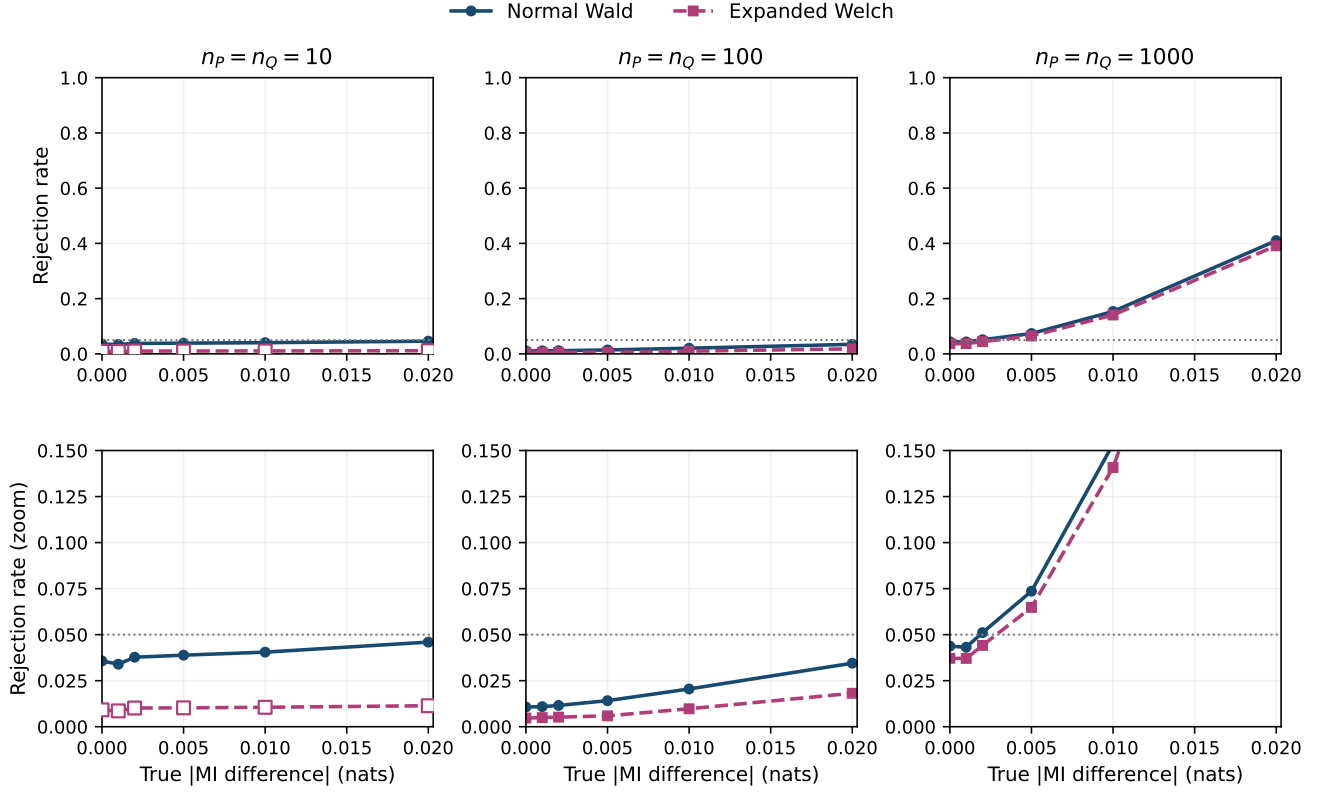
Every ordinary panel in this chapter begins at a true MI difference of zero. That point measures false-positive rejection under  $H_0$ . Points to its right measure detection under  $H_1$ . The horizontal coordinate is a *constructed population* difference in nats. The vertical coordinate is the rejection rate over 20,000 random table pairs. The blue solid circles are Normal Wald and the magenta dashed squares are Expanded Welch. The dotted horizontal line marks the nominal 0.05 rejection rate. Each plotted rate divides the number of valid rejections by all 20,000 replicates. A hollow marker indicates a valid-result rate below 0.90. The figure source manifest gives the exact saved configuration identifiers for every panel.

### 6.1 Main landscape: shape, margins and sample size

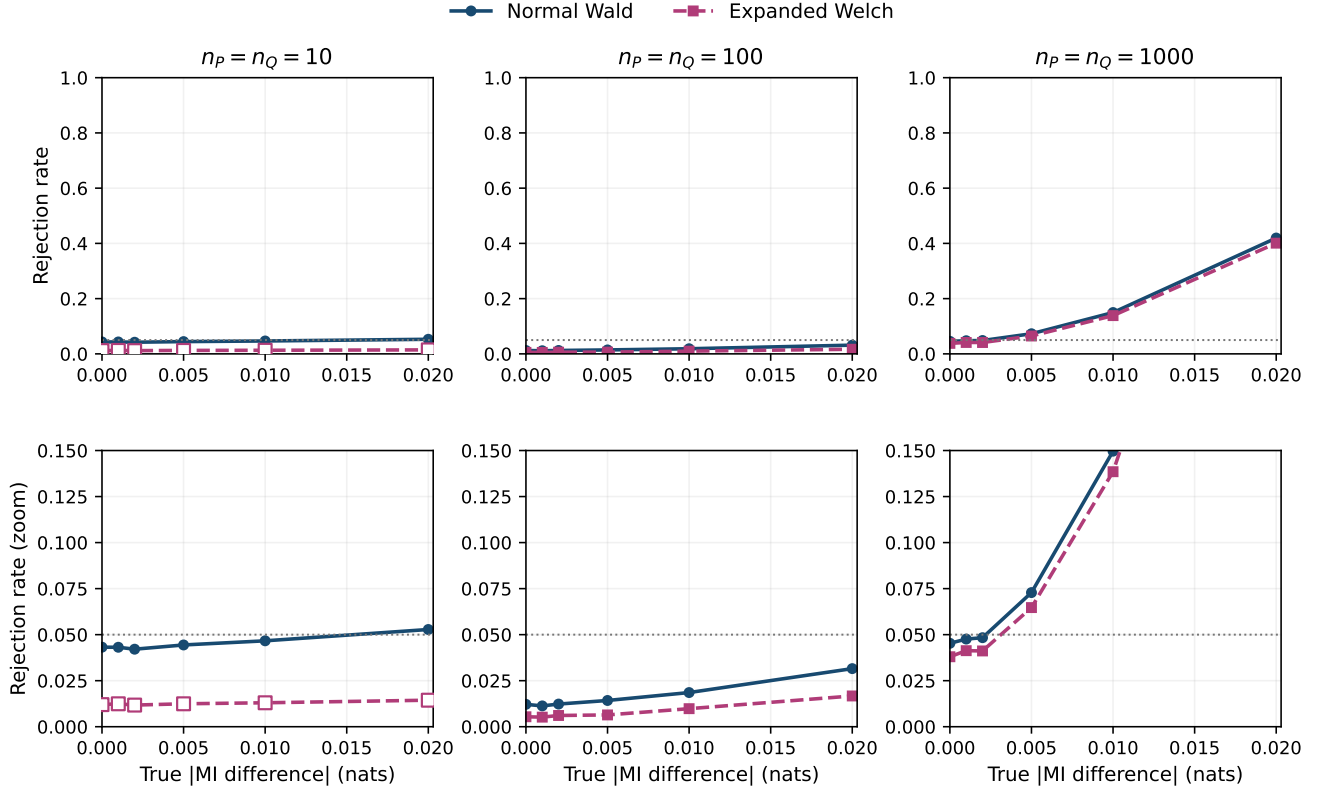
The first question is broad: do the methods behave differently as the table grows, margins become uneven and samples become larger? Figures 6.1–6.12 provide a systematic view, not a selection of favourable examples. Each of the four shapes is shown under all three margin profiles. Within every figure, the columns are the same three equal sample sizes. The full vertical scale preserves comparability of detection; the lower row enlarges the region in which calibration differences near 0.05 can be seen. The full nine-size ladder, including  $n = 2, 5, 20, 50, 250, 500$ , remains in the companion atlas.



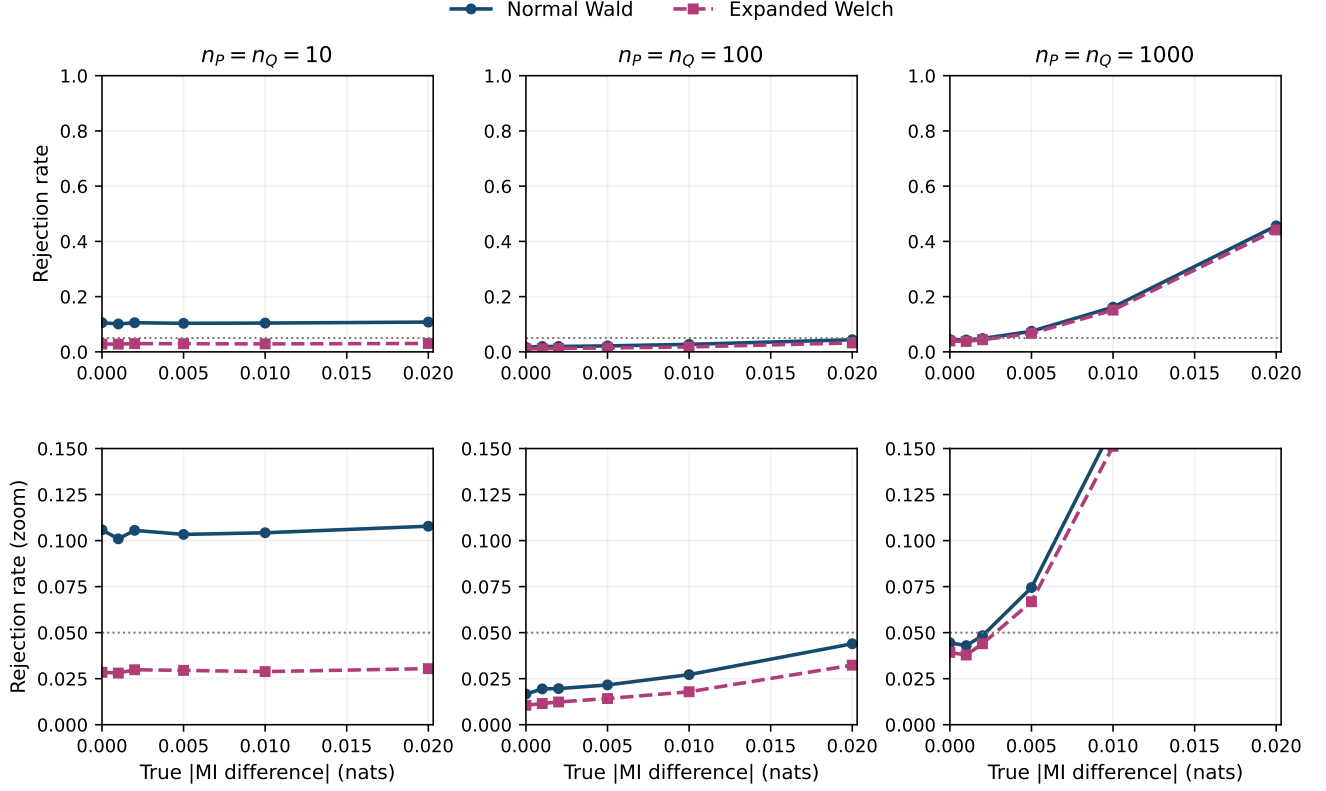
**Figure 6.1:**  $2 \times 2$  tables, both  $P$  and  $Q$  have uniform row and column margins  $(1/2, 1/2)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



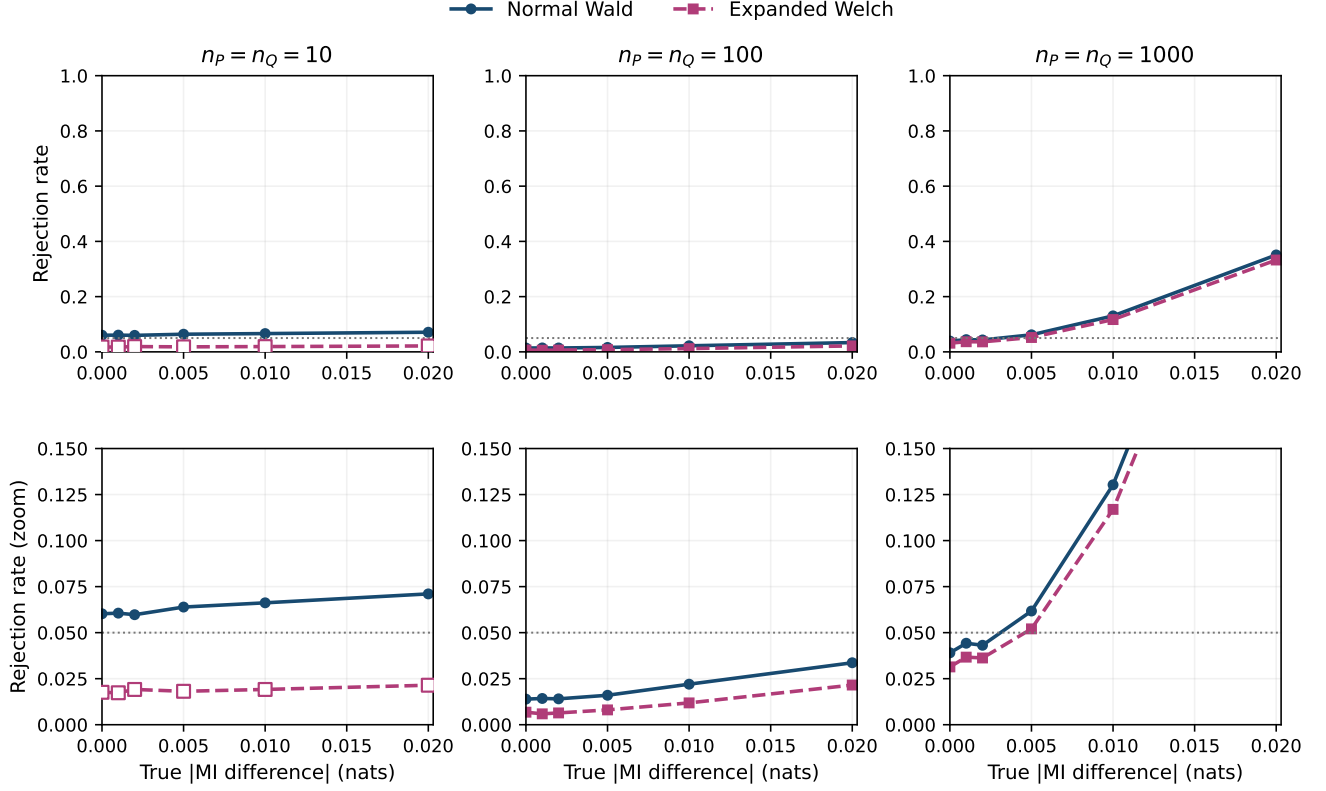
**Figure 6.2:**  $2 \times 2$  tables, both  $P$  and  $Q$  have row and column margins  $m_2(0.8) = (0.8, 0.2)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



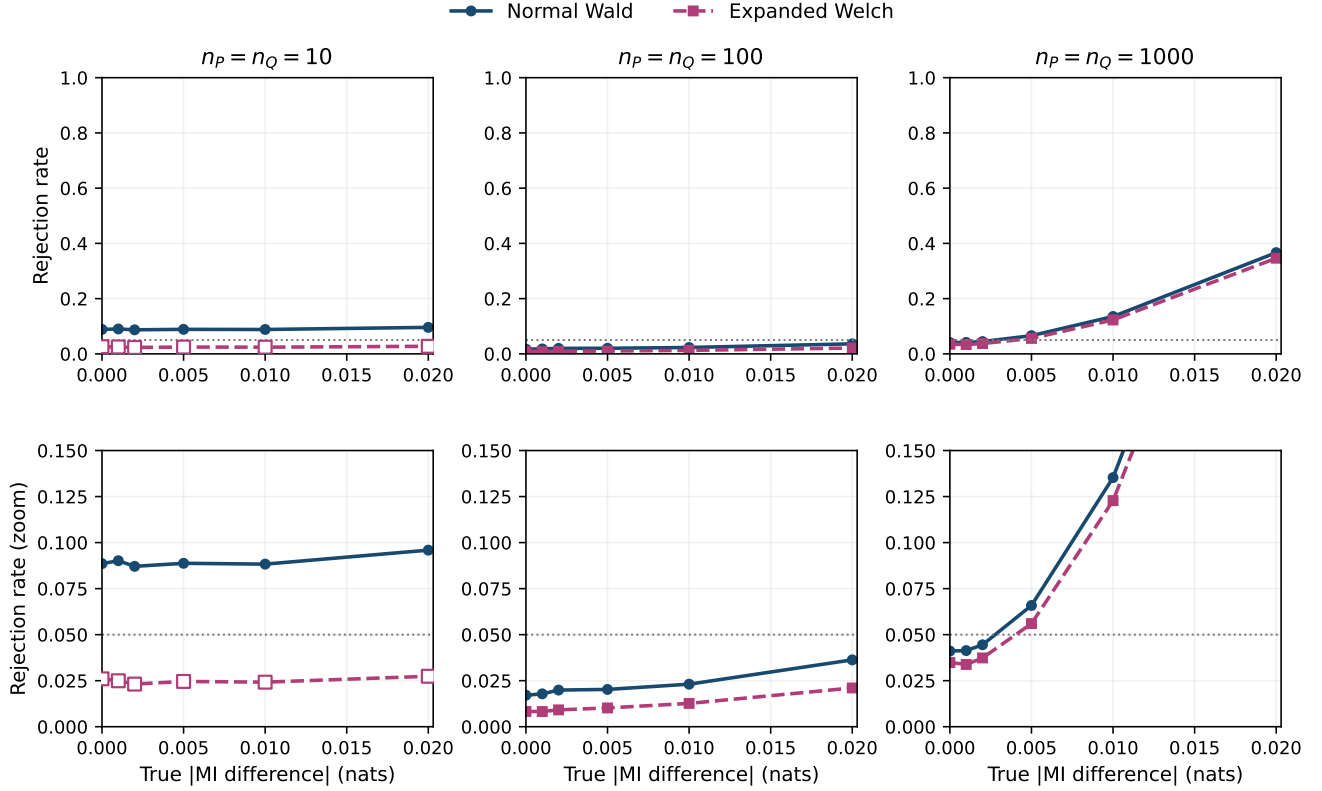
**Figure 6.3:**  $2 \times 2$  tables,  $P$  has row and column margins  $m_2(0.7) = (0.7, 0.3)$ , while  $Q$  has  $m_2(0.8) = (0.8, 0.2)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



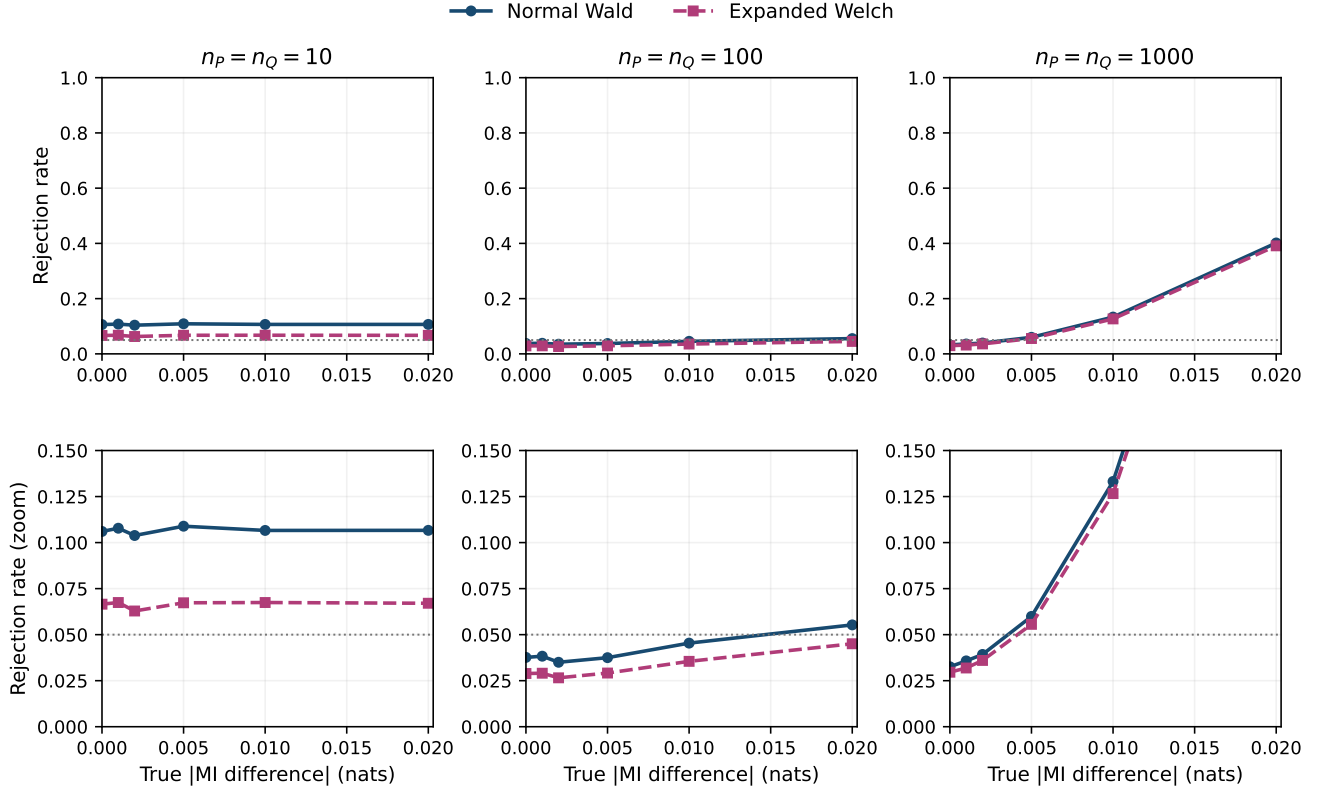
**Figure 6.4:**  $3 \times 3$  tables, both  $P$  and  $Q$  have uniform row and column margins  $(1/3, 1/3, 1/3)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



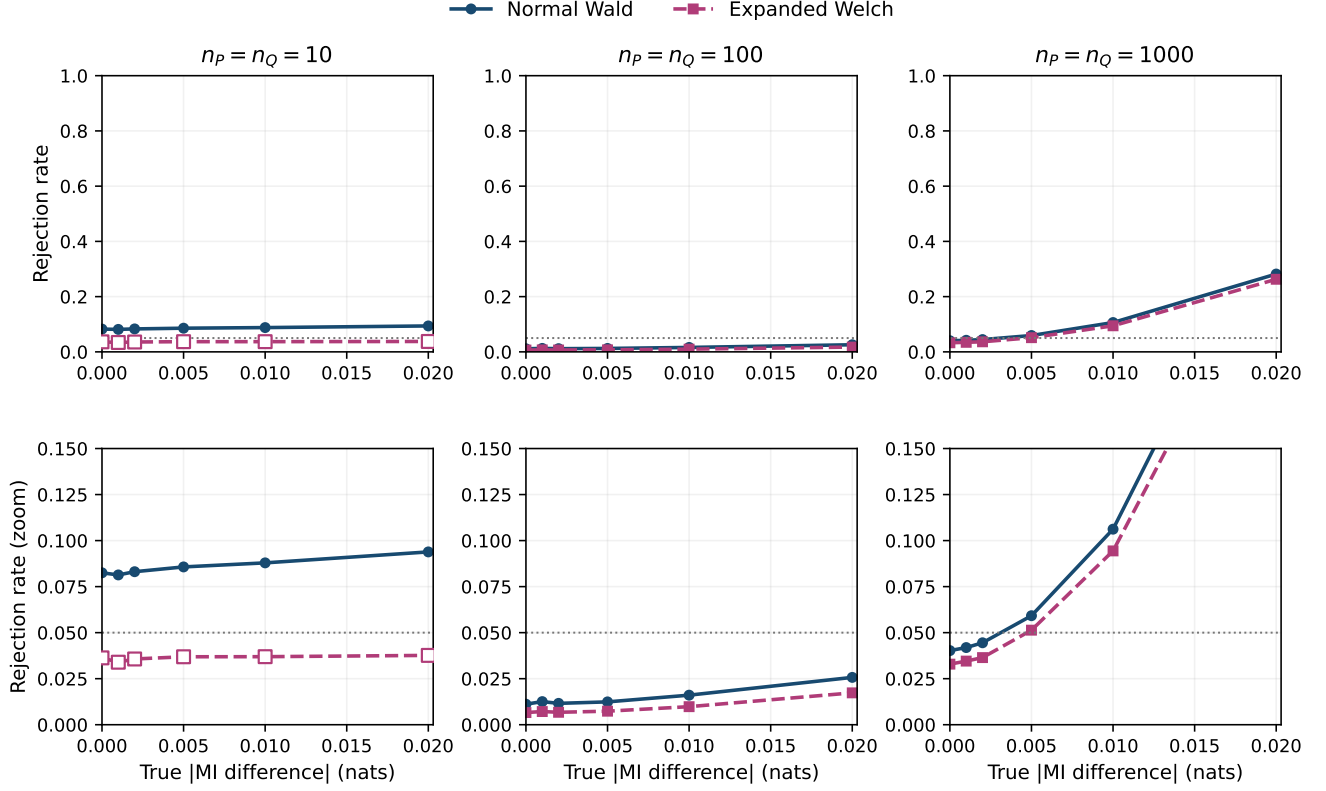
**Figure 6.5:**  $3 \times 3$  tables, both  $P$  and  $Q$  have row and column margins  $m_3(0.8) = (0.8, 0.1, 0.1)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



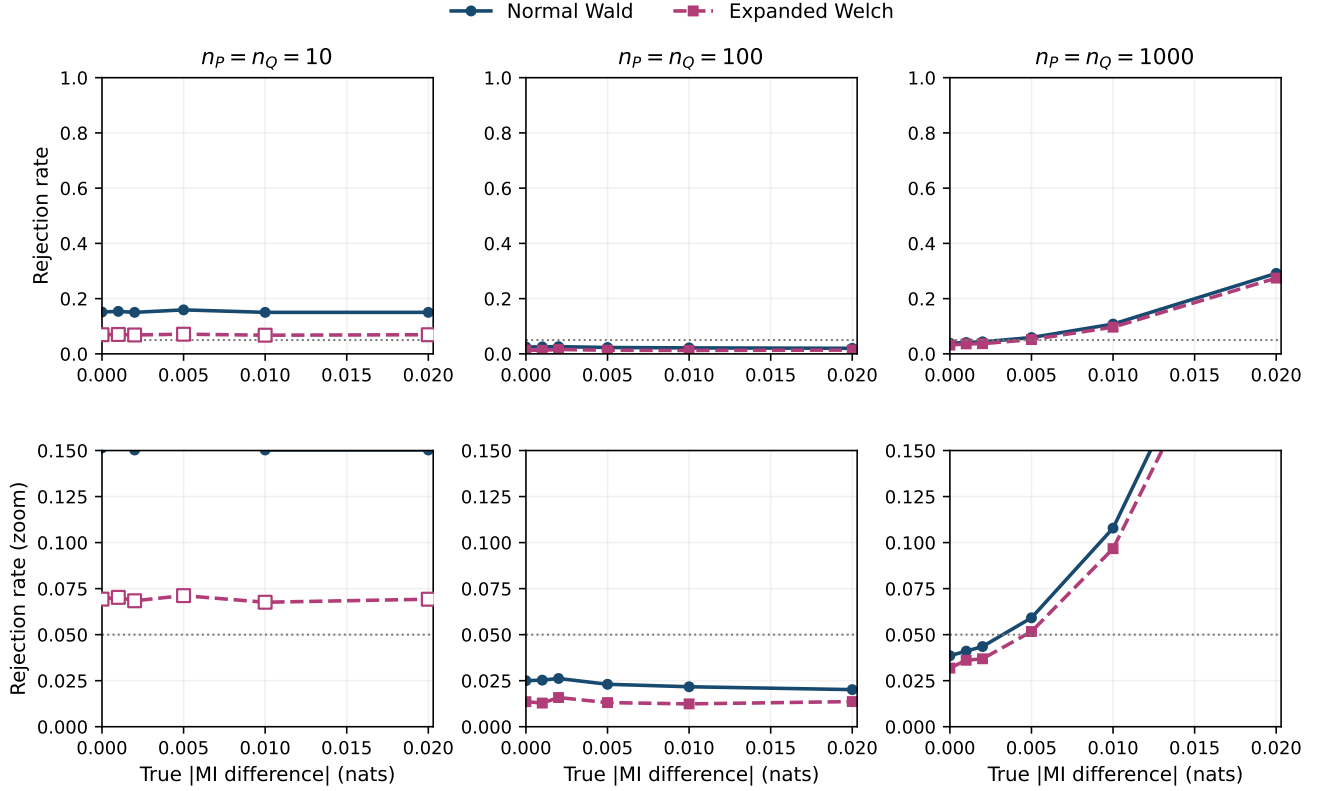
**Figure 6.6:**  $3 \times 3$  tables,  $P$  has row and column margins  $m_3(0.7) = (0.7, 0.15, 0.15)$ , while  $Q$  has  $m_3(0.8) = (0.8, 0.1, 0.1)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



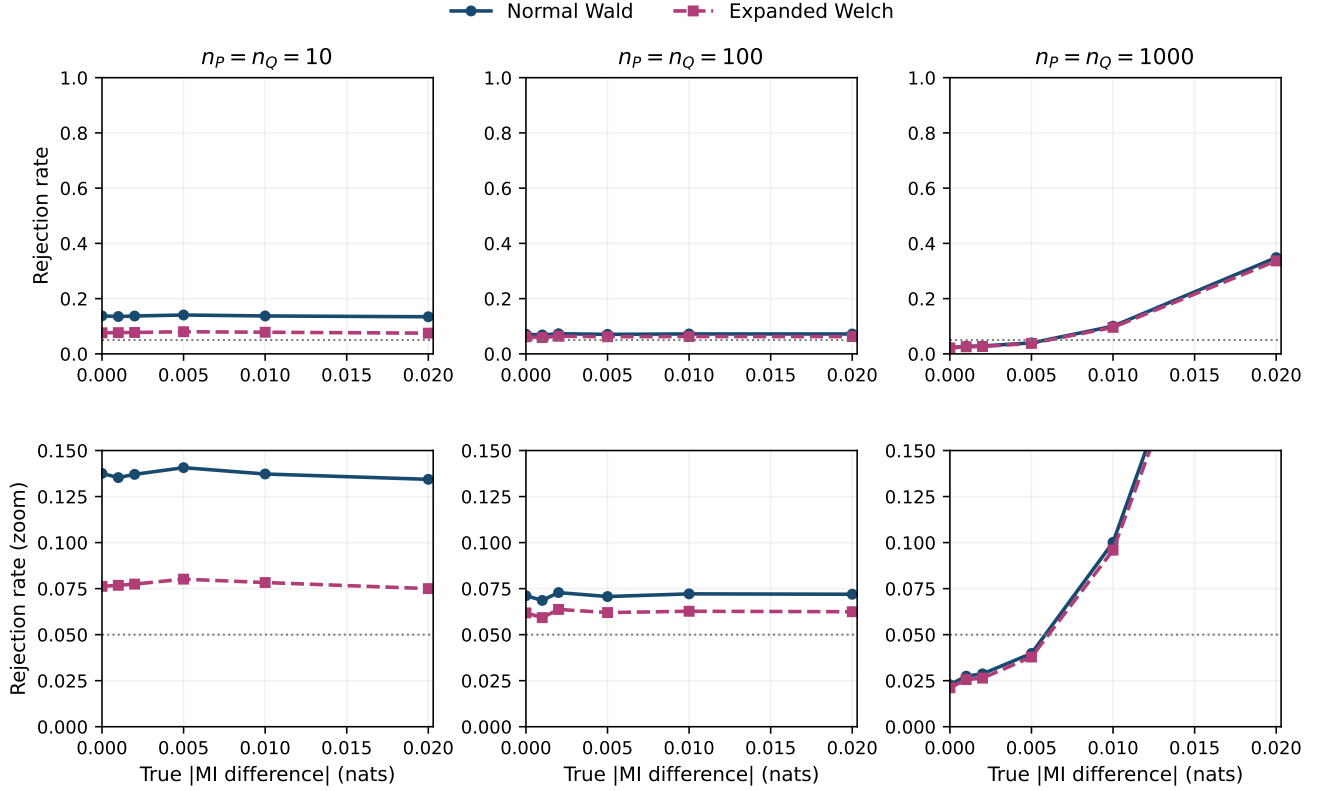
**Figure 6.7:**  $5 \times 5$  tables, both  $P$  and  $Q$  have uniform row and column margins  $(1/5, \dots, 1/5)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



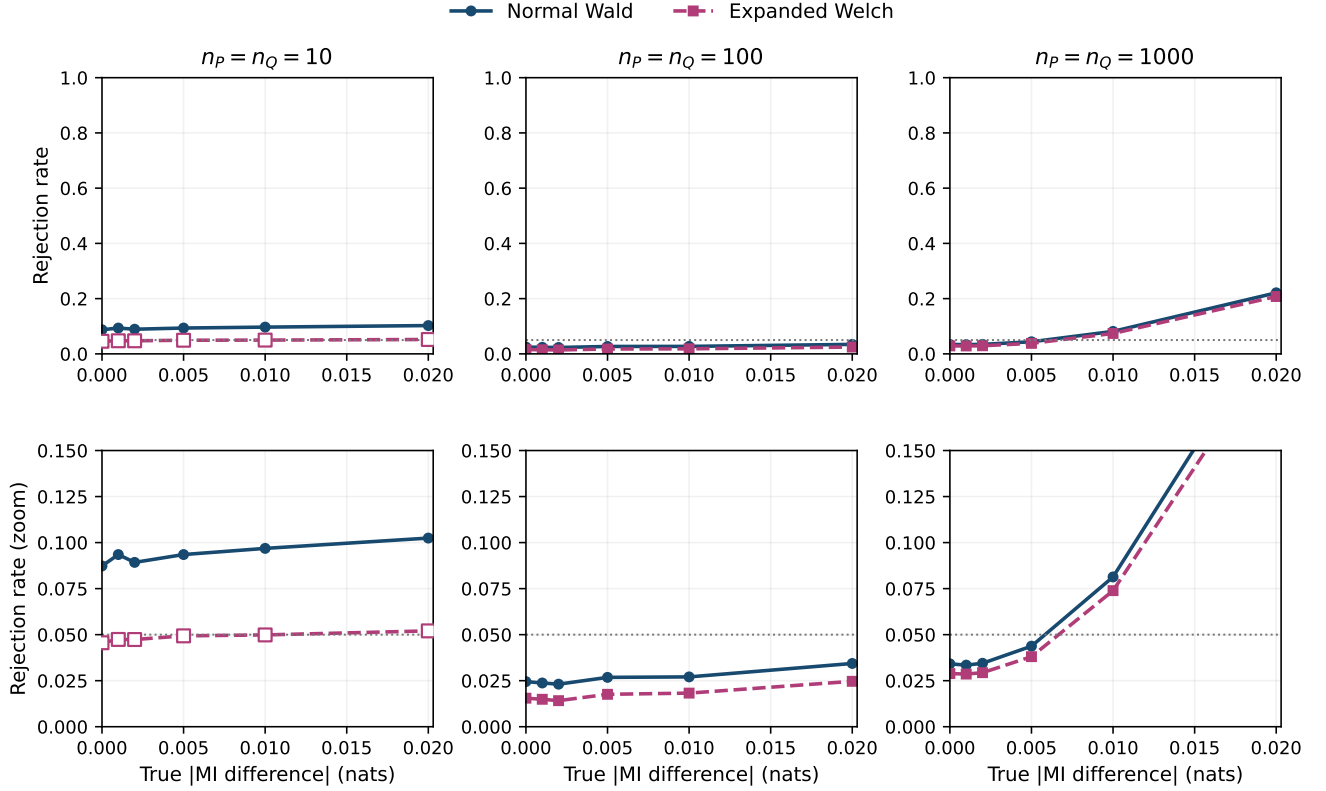
**Figure 6.8:**  $5 \times 5$  tables, both  $P$  and  $Q$  have row and column margins  $m_5(0.8) = (0.8, 0.05, 0.05, 0.05, 0.05)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



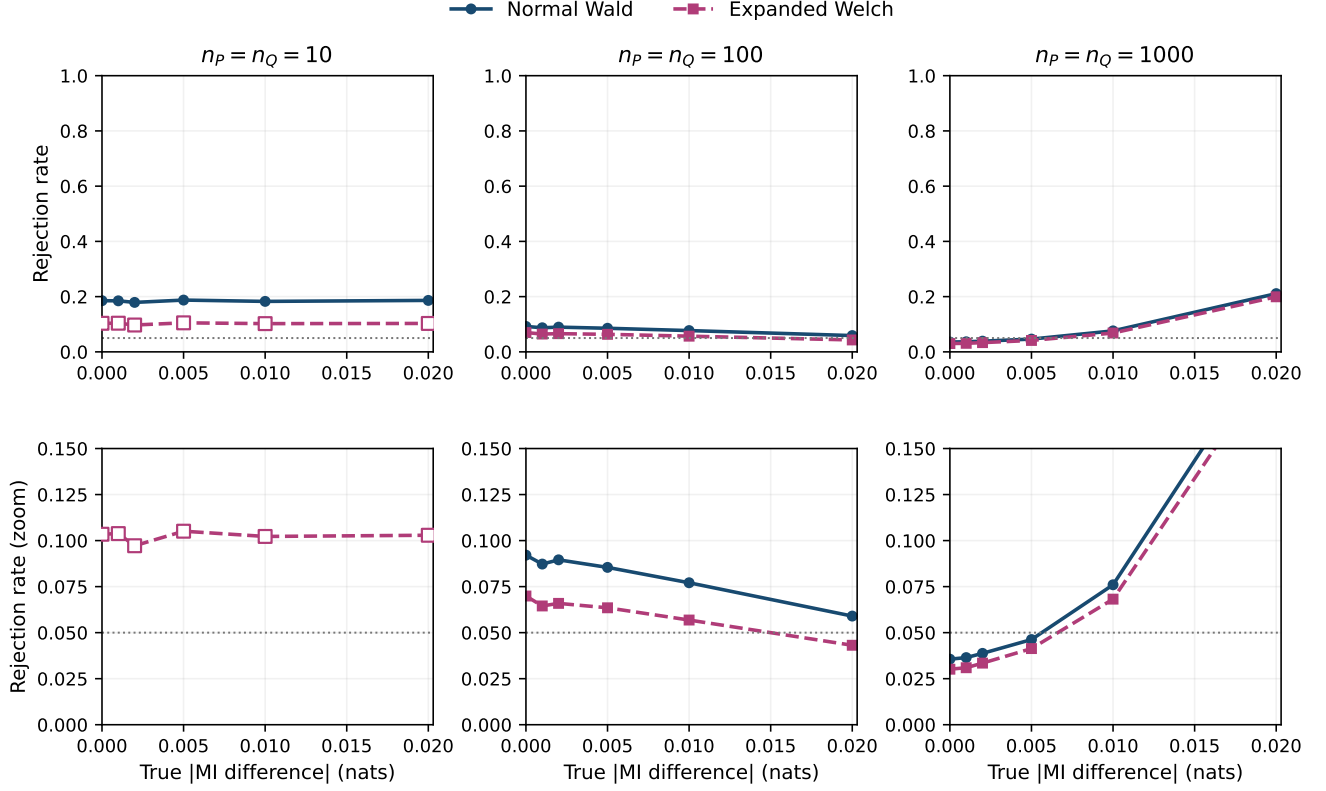
**Figure 6.9:**  $5 \times 5$  tables,  $P$  has row and column margins  $m_5(0.7) = (0.7, 0.075, 0.075, 0.075, 0.075)$ , while  $Q$  has  $m_5(0.8) = (0.8, 0.05, 0.05, 0.05, 0.05)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



**Figure 6.10:**  $8 \times 8$  tables, both  $P$  and  $Q$  have uniform row and column margins  $(1/8, \dots, 1/8)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



**Figure 6.11:**  $8 \times 8$  tables, both  $P$  and  $Q$  have row and column margins  $m_8(0.8) = (0.8, 0.2/7, \dots, 0.2/7)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



**Figure 6.12:**  $8 \times 8$  tables,  $P$  has row and column margins  $m_s(0.7) = (0.7, 0.3/7, \dots, 0.3/7)$ , while  $Q$  has  $m_s(0.8) = (0.8, 0.2/7, \dots, 0.2/7)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

The landscape does not yield a uniformly preferable method. At  $n = 5$ , Expanded Welch has a zero null rejection rate in all 12 main regimes. This includes the uniform  $3 \times 3$ ,  $5 \times 5$  and  $8 \times 8$  regimes, where more than 90% of calculations are valid, so the zero rates cannot be attributed only to invalid outputs. Among the 24 main null regimes with  $n = 10$  or 20, Expanded Welch is numerically closer to 0.05 in 15 and Wald in 9. The comparison reverses in the 24 main null settings at  $n = 500$  or 1,000: Wald is closer to 0.05 in all 24. At  $n = 1,000$ , Wald null rates across the main settings lie between 0.0228 and 0.0453; Expanded Welch lies between 0.0211 and 0.0393. Neither is uniformly at the nominal level.

Expanded Welch uses a heavier-tailed reference and can only lower rejection relative to Wald on the same table pairs. This can bring a liberal null rate toward 0.05, but moves a conservative rate farther below it. Closeness to 0.05 must therefore be judged regime by regime.

Two fully valid null points make the contrast concrete. With uniform  $8 \times 8$  populations,  $n_P = n_Q = 20$ , and equal true MI of 0.02, Wald rejects 0.06870 of samples and Expanded Welch 0.05695. The latter is closer to 0.05. With uniform  $2 \times 2$  populations at  $n_P = n_Q = 1,000$ , the respective rates are 0.04450 and 0.03865; the correction moves an already conservative procedure farther below the target. Both methods are valid on all replicates in these examples, so the change is attributable to their different references, not missing outputs.

**Table 6.1:** Two fully valid main-grid null regimes. Brackets beside each rate are pointwise Wilson 95% intervals; the final bracket is the approximate paired 95% interval for Wald minus Expanded Welch on the same samples. Source configurations are `config_313a5e5c05280f9e` and `config_700945d218841f2e`, respectively.

| Regime                             | Wald                       | Expanded                   | Wald minus<br>Expanded     |
|------------------------------------|----------------------------|----------------------------|----------------------------|
|                                    | Rate [95%<br>interval]     | Rate [95%<br>interval]     | Rate [95%<br>interval]     |
| $8 \times 8$ , uniform, $n = 20$   | 0.0687 [0.0653,<br>0.0723] | 0.0570 [0.0538,<br>0.0602] | 0.0118 [0.0103,<br>0.0132] |
| $2 \times 2$ , uniform, $n = 1000$ | 0.0445 [0.0417,<br>0.0474] | 0.0387 [0.0361,<br>0.0414] | 0.0059 [0.0048,<br>0.0069] |

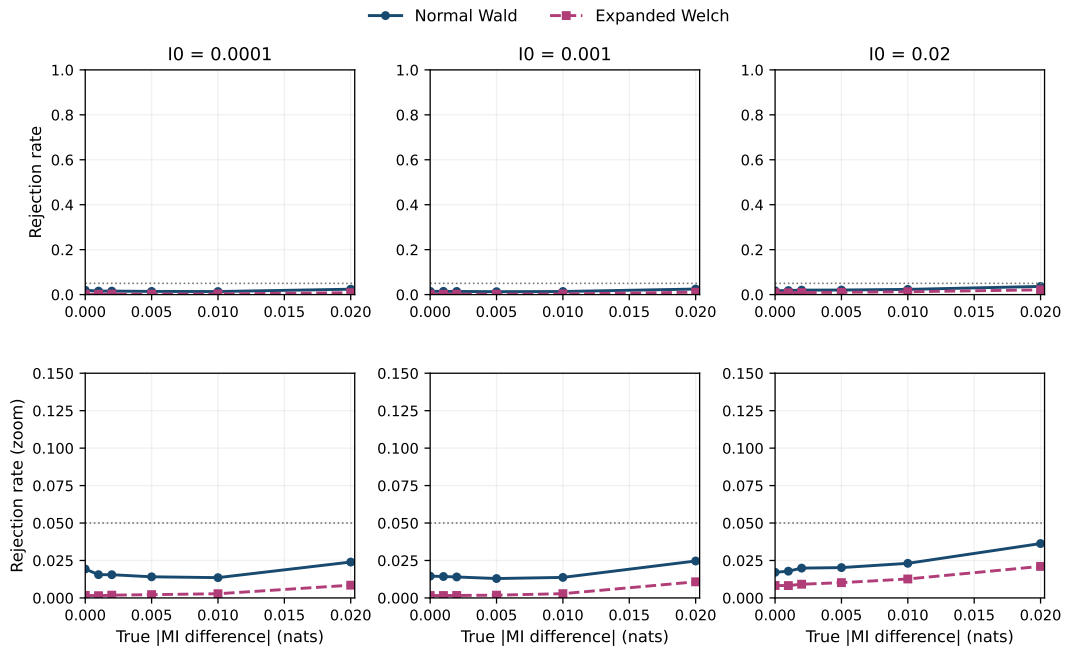
The  $n = 5$  records in the companion atlas also show an important limit to interpreting small rejection rates. In the  $8 \times 8$  different-skew alternative with  $n_P = n_Q = 5$ ,  $I(P) = 0.02$  and  $I(Q) = 0.021$ , Wald rejects 0.30865 and Expanded Welch rejects 0.00000, but their valid-result rates are only 0.83840 and 0.31415. These are probabilities of a *valid rejection*, not a clean comparison of power at matched achieved false-positive rates. The corresponding null and validity records

must be examined before calling the gap a power advantage.

## 6.2 Baseline MI and the location of changed cells

The main grid fixes  $I(P) = 0.02$ . Does starting closer to independence change the conclusion? Figure 6.13 holds shape, margins, sample size and changed cells fixed while varying only the common starting MI. The zero-difference points are especially informative: for  $I_0 = 0.0001, 0.001, 0.02$ , Wald's rates are 0.0193, 0.0146 and 0.0171, while Expanded Welch's are 0.0017, 0.0016 and 0.0083. All the plotted null calculations in this focused set are valid. Replacing the normal reference with a heavier-tailed Student reference does not address the already-low null rejection in these settings.

Starting closer to independence therefore does not create a general case for Expanded Welch. Here the main difficulty is already present in the first-order statistic and standard error; using a larger Student critical value only makes the conservative rejection rate smaller.

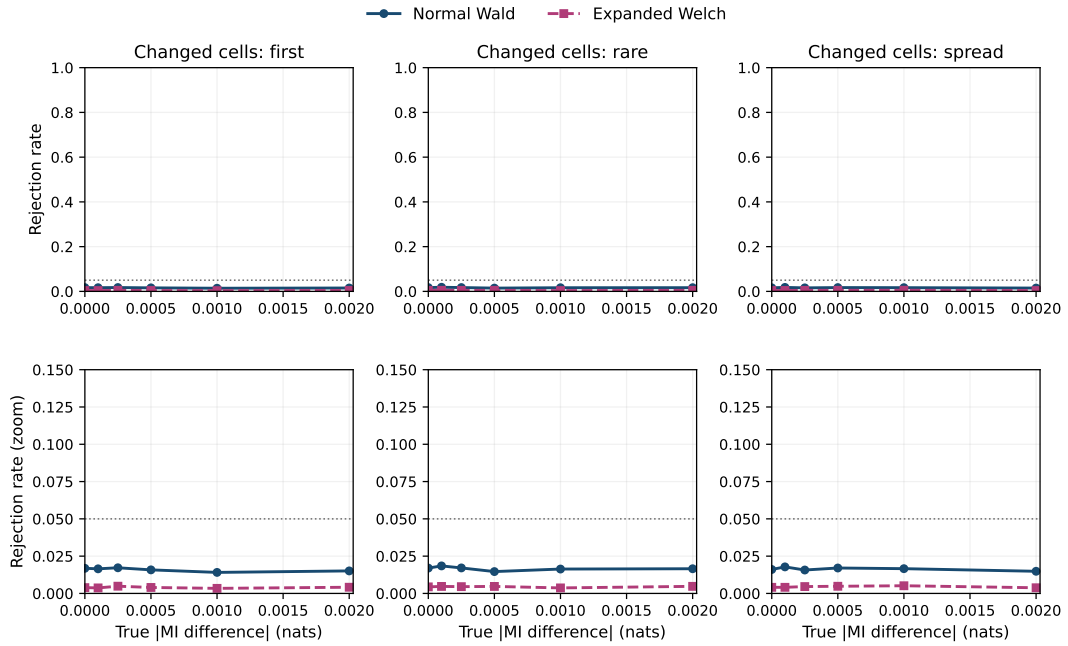


**Figure 6.13:** Baseline-MI comparison for  $3 \times 3$  tables. Columns have  $I(P) = 0.0001, 0.001, 0.02$  nats, respectively;  $I(Q) = I(P) + |\Delta|$ .  $P$  has row and column margins  $(0.7, 0.15, 0.15)$  and  $Q$  has  $(0.8, 0.1, 0.1)$ . Both use the first-cell change block,  $n_P = n_Q = 100$ , and  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1 rejection scale; bottom: 0–0.15 zoom. Each point has 20,000 replicates; hollow markers flag validity below 90%.

Figure 6.14 asks whether the association being placed in an early, rare or distributed set of cells changes performance when the margins and target MI are

fixed. At the displayed null points, Wald rates are 0.0169, 0.0170 and 0.0160 and Expanded Welch rates are 0.0039, 0.0044 and 0.0041 for first, rare and spread patterns. The similarity of those three null rates in this selected setting is not evidence that the patterns are equivalent in all tables. The full  $3 \times 3$ ,  $5 \times 5$  and  $8 \times 8$  matched sets are in the atlas.

Within this controlled comparison, moving the association to different cells does not remove the conservative gap between the methods. The implication is limited but useful: the displayed result is not an artefact of only the first  $2 \times 2$  change block used in the main landscape.



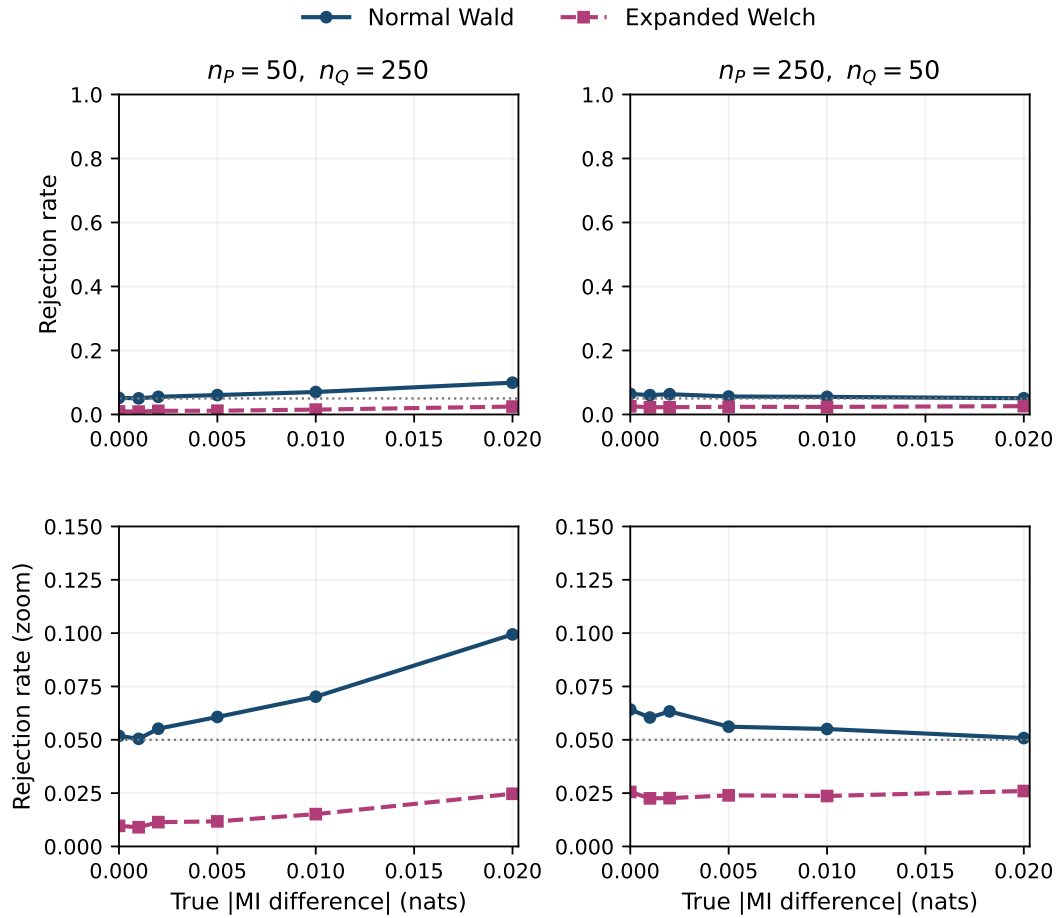
**Figure 6.14:** Changed-cell comparison for  $5 \times 5$  tables. Columns place association in the first  $2 \times 2$  block, last (rare)  $2 \times 2$  block, or a centred score outer product spread across all cells. For every column,  $P$ 's row and column margins are  $(0.7, 0.075, 0.075, 0.075, 0.075)$ , and  $Q$ 's are  $(0.8, 0.05, 0.05, 0.05, 0.05)$ . Here  $I(P) = 0.0001$ ,  $I(Q) = I(P) + |\Delta|$ ,  $n_P = n_Q = 100$ , and  $|\Delta| = 0, 0.0001, 0.00025, 0.0005, 0.001, 0.002$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

## 6.3 Unequal samples and the direction of the difference

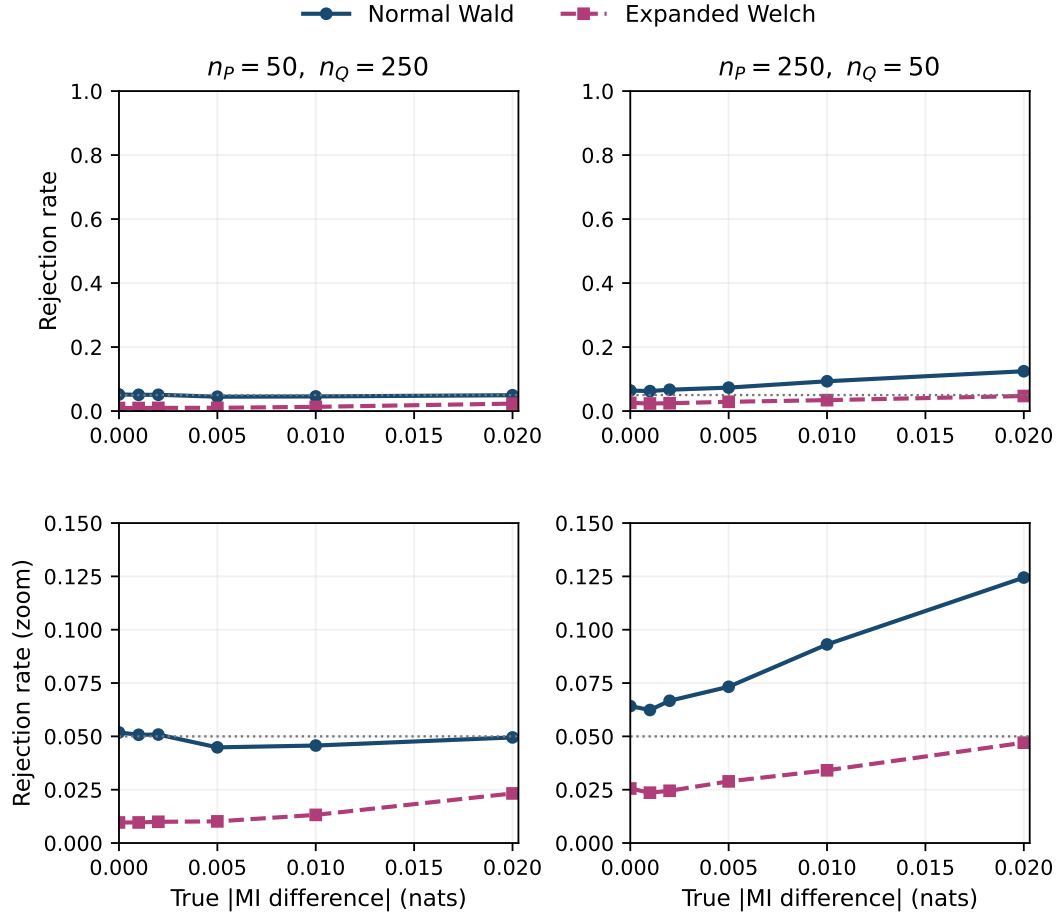
Figures 6.15 and 6.16 hold the unordered sample sizes at 50 and 250 but reverse which population has the larger sample. They also show the two directions of the MI difference separately. The distinction matters because the finite bias corrections are  $(r-1)(c-1)/(2n_P)$  and  $(r-1)(c-1)/(2n_Q)$ , and the two populations have

different margins. For example, at  $(n_P, n_Q) = (50, 250)$  and a 0.02-nat difference, Wald's rejection is 0.0994 when  $Q$  has higher MI and 0.0495 when  $P$  has higher MI. Expanded Welch gives 0.0247 and 0.0232. All four rates in that specific comparison have validity one. These alternative rates should still be read beside the matched null points; they are not size-adjusted power.

The practical implication is that an absolute MI difference does not fully specify the testing problem. Which population has the higher MI and which one has the larger sample changes the bias and variance contributions. Power should therefore not be averaged across the two directions of sample imbalance.



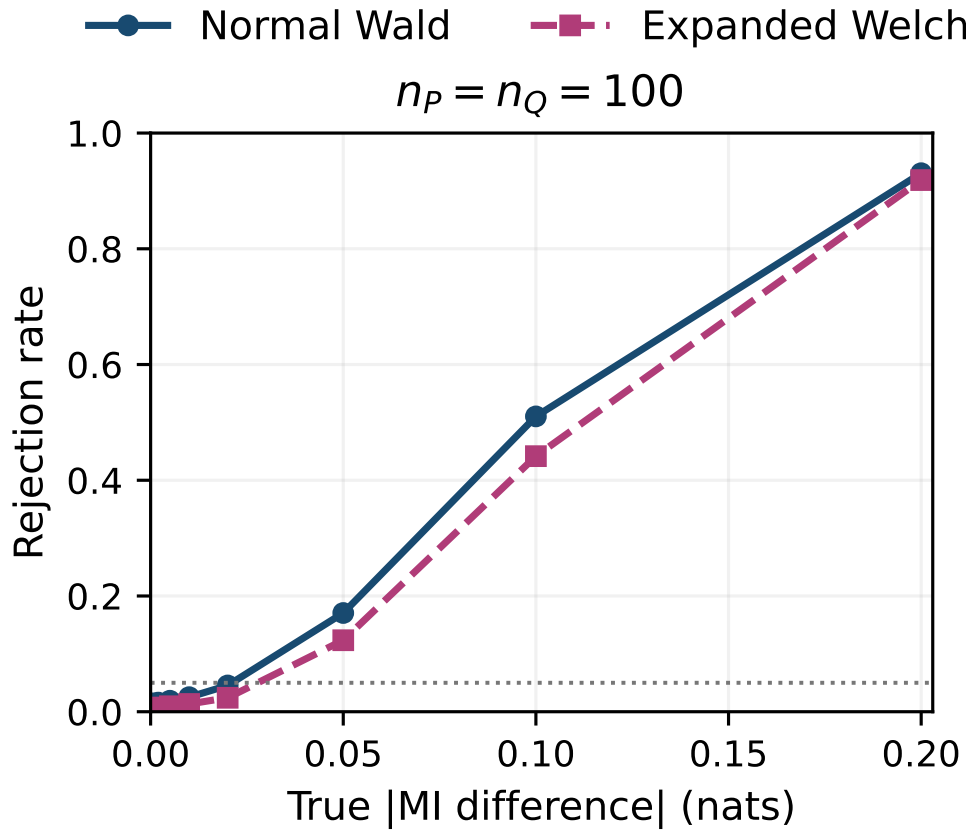
**Figure 6.15:** Unequal samples,  $Q$  higher. Columns have  $(n_P, n_Q) = (50, 250)$  and  $(250, 50)$ . For both columns,  $P$  has  $3 \times 3$  row and column margins  $(0.7, 0.15, 0.15)$  and  $Q$  has  $(0.8, 0.1, 0.1)$ ; the first-cell change block is used.  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ , with  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.



**Figure 6.16:** Unequal samples,  $P$  higher. Columns have  $(n_P, n_Q) = (50, 250)$  and  $(250, 50)$ .  $P$ 's  $3 \times 3$  row and column margins are  $(0.7, 0.15, 0.15)$ ;  $Q$ 's are  $(0.8, 0.1, 0.1)$ . Both use the first-cell change block.  $I(Q) = 0.02$  and  $I(P) = 0.02 + |\Delta|$ , with  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

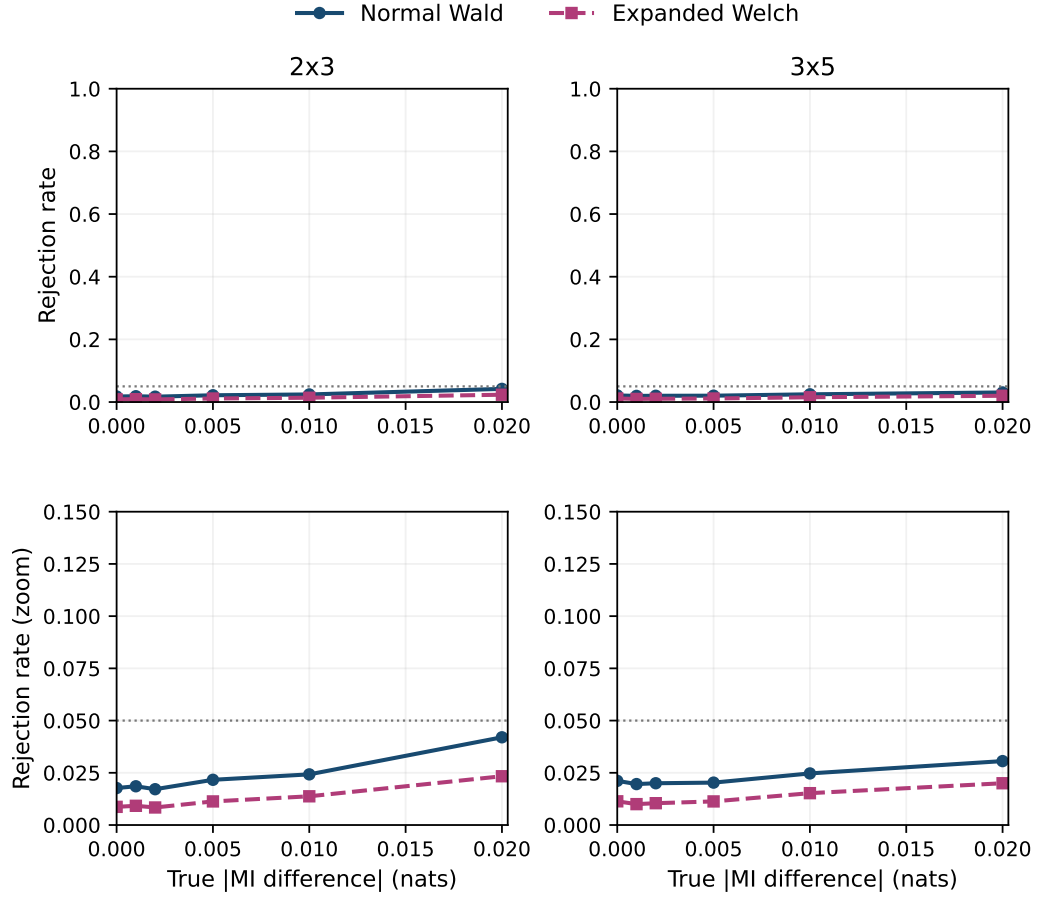
## 6.4 Larger differences and rectangular tables

The main grid deliberately concentrates on small MI differences. A separate  $2 \times 2$  family extends the feasible axis to 0.2 nats without changing the axis units, using only targets the construction can reach. This shows where detection eventually rises, while retaining the zero-difference point that defines null calibration. Rectangular  $2 \times 3$  and  $3 \times 5$  tables check that the discussion is not tied only to square alphabets. The panels do not average over shapes.



**Figure 6.17:** Wider MI differences in a uniform  $2 \times 2$  population pair. The single panel has  $P$  and  $Q$  row and column margins  $(0.5, 0.5)$ , first-cell changes,  $n_P = n_Q = 100$ ,  $I(P) = 0.02$ , and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2$  nats; the rejection axis is 0–1. Each point has 20,000 replicates.

The wider  $2 \times 2$  curve shows that both methods eventually detect large MI differences, while Expanded Welch remains below Wald until the rejection rates approach one. This is the expected cost of its larger critical value: the power loss matters most at intermediate differences and becomes small when the signal is overwhelming.

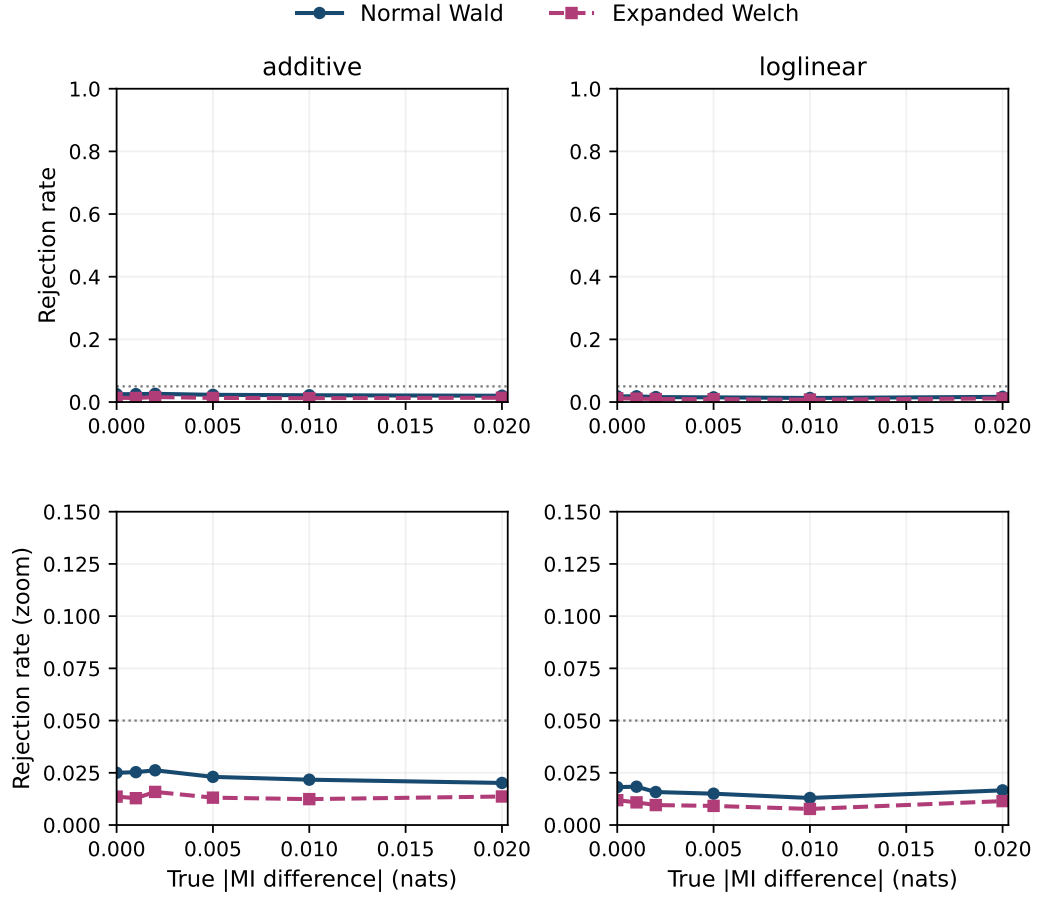


**Figure 6.18:** Rectangular tables. Columns are  $2 \times 3$  and  $3 \times 5$ . In each dimension  $k$ ,  $P$ 's corresponding margin is  $m_k(0.7)$  and  $Q$ 's is  $m_k(0.8)$ , for both rows and columns. Thus the first column uses  $P$  rows  $(0.7, 0.3)$ , columns  $(0.7, 0.15, 0.15)$ , and  $Q$  rows  $(0.8, 0.2)$ , columns  $(0.8, 0.1, 0.1)$ ; the second applies the same formula to 3 and 5 categories. Both use first-cell changes,  $n_P = n_Q = 100$ ,  $I(P) = 0.02$ , and  $I(Q) = 0.02 + |\Delta|$  for  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

In the rectangular tables, both methods remain below 0.05 throughout the displayed range, with Expanded Welch lower still. Those low alternative rejection rates indicate weak detection. The rectangular tables therefore show the conservative half of the trade-off, while the wider  $2 \times 2$  curve shows where detection eventually becomes strong.

## 6.5 Comparing Two Population Constructions

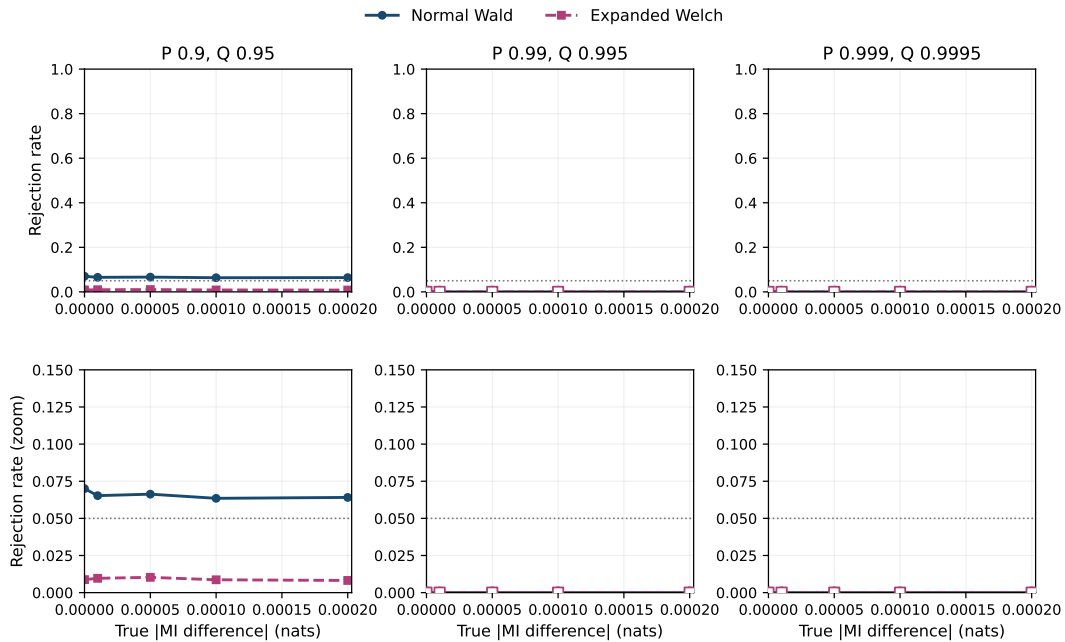
The additive block is interpretable, but it is only one way to make tables with fixed margins and target MI. Figure 6.19 compares it with an ordinal log-linear construction under matched targets. The null rates are not identical: for the displayed  $5 \times 5$  different-skew comparison at  $n = 100$ , Wald gives 0.0250 under additive construction and 0.0181 under log-linear construction; Expanded Welch gives 0.0136 and 0.0119. All are below nominal and all are valid. Thus matching the scalar MI and margins does not fix the full finite-sample behaviour. The constructor comparison provides a sensitivity check across two dependence structures.



**Figure 6.19:** Additive versus ordinal log-linear construction for  $5 \times 5$  tables. The left column uses the first-cell additive block; the right fits an ordinal interaction while preserving the margins. In both,  $P$ 's row and column margins are  $(0.7, 0.075, 0.075, 0.075, 0.075)$  and  $Q$ 's are  $(0.8, 0.05, 0.05, 0.05, 0.05)$ . Both have  $I(P) = 0.02$ ,  $I(Q) = 0.02 + |\Delta|$ ,  $n_P = n_Q = 100$ , and  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

## 6.6 Extreme skew, independence and larger samples

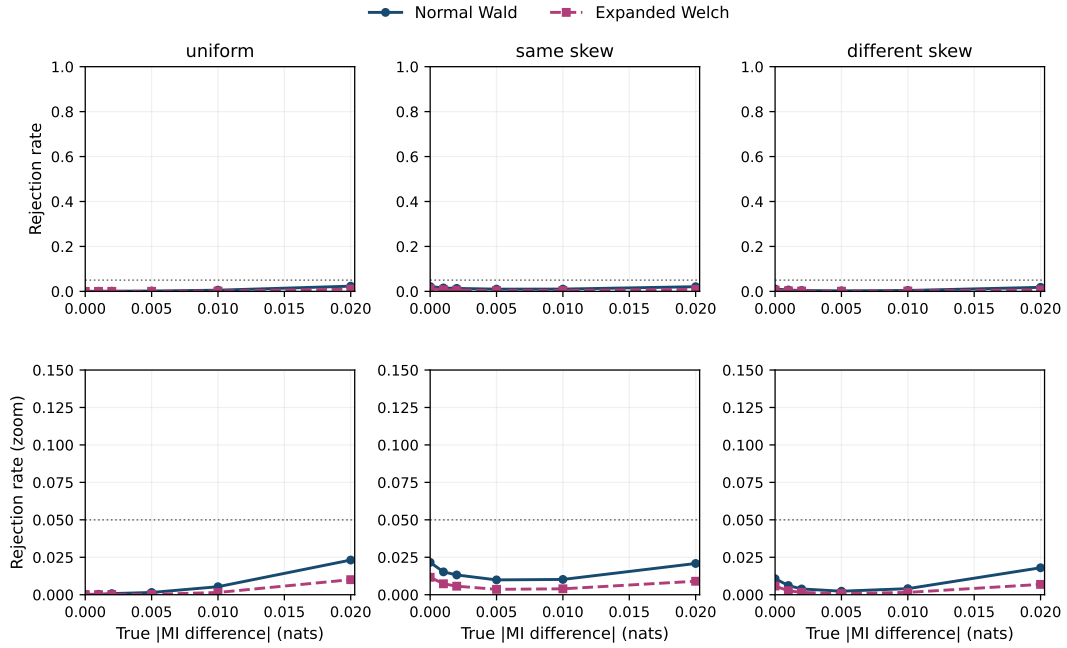
The extreme-skew check deliberately keeps configurations with very small expected cell counts. At  $n_P = n_Q = 100$ ,  $3 \times 3$  tables with dominant margins 0.999 for  $P$  and 0.9995 for  $Q$  yield valid null results in only 0.0118 of Wald replicates and 0.00005 of Expanded Welch replicates under the strict component-validity rule. Both unconditional false-positive rates are zero, but this is mostly an absence of valid tests, not evidence of excellent calibration. With dominant margins 0.9 and 0.95, the corresponding null rates are 0.0701 and 0.0088; Expanded Welch lowers an excessive Wald rate past the target. Figure 6.20 compares all three specified skew levels.



**Figure 6.20:** Extreme-skew  $3 \times 3$  tables. Across columns the dominant row and column probabilities for  $(P, Q)$  are  $(0.9, 0.95)$ ,  $(0.99, 0.995)$ , and  $(0.999, 0.9995)$ . Each remaining margin entry splits the leftover probability equally between the two other categories. Both use first-cell changes,  $n_P = n_Q = 100$ ,  $I(P) = 0.00001$ , and  $I(Q) = I(P) + |\Delta|$ , for  $|\Delta| = 0, 0.00001, 0.00005, 0.0001, 0.0002$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates; hollow markers flag validity below 90%.

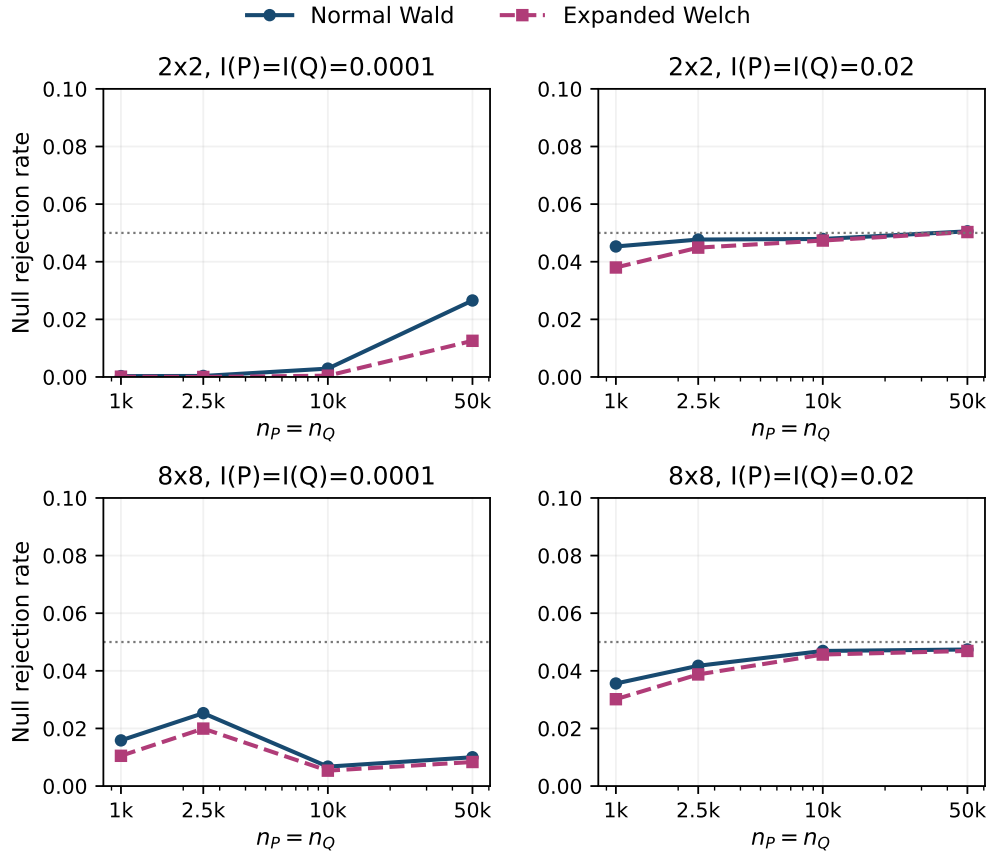
Exact independence is a separate theoretical boundary. Figure 6.21 starts  $P$  at MI zero and includes the equal-MI point where both populations are independent. This is not a routine positive-MI validation of Expanded Welch: the population first-order MI variance is zero at independence. At  $n = 100$ , for example, the uniform  $2 \times 2$  null rates are only 0.00025 and 0.00010 for Wald and Expanded Welch. Appendix D explains why a second-order independence test has a different

reference.



**Figure 6.21:** Exact-independence boundary for  $2 \times 2$  tables. Columns are uniform margins (0.5, 0.5) for both populations, same skew (0.8, 0.2) for both, and different skew with  $P = (0.7, 0.3)$  and  $Q = (0.8, 0.2)$ ; each vector is used for both rows and columns. The first-cell block is used,  $I(P) = 0$ ,  $I(Q) = |\Delta|$ ,  $n_P = n_Q = 100$ , and  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

Does increasing  $n$  resolve the null error when the populations remain fixed? Figure 6.22 keeps equal MI and changes only sample size. Across the 12 first-block shape–margin combinations with  $I(P) = I(Q) = 0.02$  and  $n = 50,000$ , Wald’s false-positive rates range from 0.0470 to 0.0527 and Expanded Welch’s from 0.0466 to 0.0526. At equal MI 0.0001, however, the ranges at the same  $n$  are 0.0089–0.0280 and 0.0076–0.0138. These fixed-population results show that convergence can remain slow when the true MI is very small.



**Figure 6.22:** Large-sample null behaviour. Rows compare  $2 \times 2$  and  $8 \times 8$  tables; columns have common true MI 0.0001 and 0.02 nats. For both shapes,  $P$ 's row and column margins are  $m_k(0.7)$  and  $Q$ 's are  $m_k(0.8)$ , with  $k = 2$  or  $8$ . Both populations use the first-cell block. The horizontal axis is the equal sample size  $n_P = n_Q = 1,000, 2,500, 10,000, 50,000$  on a labelled log scale; the vertical axis is null rejection from 0 to 0.10. Each point has 20,000 replicates.

## 6.7 Null Rejection Rates and the Standard-Error Check

The main-grid null contains two cases distinguished by whether the population tables themselves are equal. The 72 uniform and same-skew configurations have  $P = Q$ , whereas the 36 different-skew configurations have  $P \neq Q$  but  $I(P) = I(Q)$ . These cases, where the tables differ but their MI values agree, most directly match the thesis motivation. Table 6.2 reports these profiles separately. The broad interval 0.025–0.075 is used only to describe the landscape; it is not a formal acceptance region for calibration.

**Table 6.2:** Location, separately for each method, of the 108 main-grid null configurations relative to the descriptive rejection-rate interval 0.025–0.075. Each profile contains 36 configurations. The final column separately counts configurations with validity below 90%; it overlaps the three rate-location columns. In particular, all four  $n = 2$  configurations per profile have zero validity for both methods. Uniform and same-skew have  $P = Q$ ; different-skew has  $P \neq Q$  but  $I(P) = I(Q)$ . Exact rates and validity are available in the saved results described in Appendix C.

| Margins        | Method         | Below | Inside | Above | Validity below 90% |
|----------------|----------------|-------|--------|-------|--------------------|
| Uniform        | Normal Wald    | 10    | 18     | 8     | 4                  |
| Uniform        | Expanded Welch | 19    | 16     | 1     | 5                  |
| Same skew      | Normal Wald    | 14    | 15     | 7     | 8                  |
| Same skew      | Expanded Welch | 25    | 11     | 0     | 12                 |
| Different skew | Normal Wald    | 9     | 16     | 11    | 8                  |
| Different skew | Expanded Welch | 19    | 13     | 4     | 12                 |

The table makes the direction of the adjustment clear. Expanded Welch greatly reduces the number of rates above the descriptive band, but it also increases the number below it in every margin profile. Its apparent protection against liberal rejection is therefore purchased with additional conservatism rather than with calibration in both directions.

A separate supplementary study tested the mechanism without altering the confirmatory results. For the representative  $3 \times 3$  different-skew null with  $I(P) = I(Q) = 0.02$  and  $n = 100$ , ordinary Wald divided each observed bias-corrected MI difference by its own plug-in standard error and rejected 0.0166 of sample pairs. Keeping those same differences but dividing each by one fixed finite-sample standard deviation estimated from an independent pilot gave a rejection rate of 0.0551. Dividing instead by the fixed population first-order standard deviation gave 0.0976. The confirmatory Wald rate was 0.0171; its small difference from 0.0166 is Monte Carlo variation between independent runs.

Across the evaluated pairs, the average squared plug-in standard error was about 1.17 times the finite-sample variance of the MI difference estimated from the independent pilot. The population first-order variance was smaller than that finite-sample variance. These differences help explain why the three denominators produce different rejection rates. The two fixed denominators are diagnostics, not proposed tests: Expanded Welch changes neither the observed MI differences nor their plug-in standard errors. Full pilot and method-comparison results are retained in the experiment supplement.

## 6.8 Runtime and summary of evidence

The full method calls have the same  $O(rc)$  asymptotic complexity. The observed median time across the 60 timing regimes is about 85 microseconds for Wald and 150 microseconds for Expanded Welch. The median Expanded-to-Wald ratio is about 1.75, with regime ratios about 1.72–1.79. For the explicitly timed  $3 \times 3$ , different-skew,  $n_P = n_Q = 100$ , equal-MI input regime, the medians are 85.083 and 149.979 microseconds. These are implementation- and machine-specific timings, not universal complexity constants. Their near-flat values mainly measure fixed per-call overhead and do not empirically establish the  $O(rc)$  scaling, which follows instead from the operation count in Section 4.4. Exact per-regime medians and interquartile ranges are in the saved runtime file cited in Appendix C.

Computationally, the correction is feasible: it has the same cell-linear order as Wald and adds roughly 75% to a scalar call in this implementation. The overhead is modest compared with resampling, but it is still an extra cost. Because the statistical results do not show a general improvement, runtime does not provide a separate reason to prefer Expanded Welch.

**Table 6.3:** Scalar-call times in microseconds for the different-skew null at  $n_P = n_Q = 100$  and  $I(P) = I(Q) = 0.02$ . Entries give median [interquartile range] over five passes on 200 saved inputs per regime. Population construction and multinomial sampling are excluded.

| Shape        | Normal Wald       | Expanded Welch       | Expanded/Wald |
|--------------|-------------------|----------------------|---------------|
| $2 \times 2$ | 85.8 [85.5, 86.3] | 150.0 [149.4, 151.3] | 1.75          |
| $3 \times 3$ | 85.1 [84.6, 85.6] | 150.0 [148.9, 150.9] | 1.76          |
| $3 \times 5$ | 85.2 [84.9, 85.5] | 150.1 [149.5, 150.9] | 1.76          |
| $5 \times 5$ | 85.3 [85.1, 85.7] | 149.2 [148.5, 150.1] | 1.75          |
| $8 \times 8$ | 88.0 [87.1, 88.9] | 153.4 [152.4, 154.8] | 1.74          |

The evidence gives three conclusions. Expanded Welch can move a liberal Wald null rate closer to 0.05, but its heavier-tailed reference also reduces rejection where

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Wald is already conservative. In sparse samples, the unconditional rate can fall because Expanded Welch is more often invalid under the strict component-validity rule. Some failures would not occur under the direct combined formula and are not needed for the first two findings. These behaviours cannot be summarised by one number; the matched curves and validity records underpin the discussion in Chapter 7.

# Chapter 7

## Discussion

The proposed correction models how the estimated MI variance changes across samples. The experiments show why this alone is insufficient: Wald and Expanded Welch use the same observed MI difference and plug-in standard error. Changing the reference distribution leaves any finite-sample error in those quantities untouched.

### 7.1 Answers to the research questions

The first question was how to choose degrees of freedom for an MI test. Chapter 4 shows how one extra observation changes the estimated MI variance. That observation changes not only its cell but also the row and column totals used in other MI calculations. Measuring how much the variance estimate changes provides the degrees of freedom used by Expanded Welch.

The second question concerned false positives, detection and usable results. Table 6.1 shows both sides of the trade-off. In one case Wald rejects equal MI too often, and Expanded Welch moves the rate toward 0.05. In another, Wald already rejects too rarely, and the same adjustment moves the rate farther from 0.05. In very sparse settings, some low rejection rates arise because the method fails to return a p-value. Thus a smaller false-positive rate in one case is not evidence that Expanded Welch is better overall. Under unequal MI, it usually detects fewer differences, although the methods are not being compared at matched false-positive rates.

The third question asked which population features matter. The results vary table size, uneven probabilities, starting MI, the cells carrying the association, and the allocation of samples between the groups. These features can change rejection rates even when the true MI difference is held fixed. In particular, tables with the same row totals, column totals and MI can still have different cell probabilities and

behave differently when sampled.

The fourth question concerned larger samples and computation. For the tested populations with MI 0.02, both methods reject equal MI at close to the target 5% rate when each sample contains 50,000 observations. They still reject too rarely for the tested populations with very small MI. In the measured implementation, one Expanded Welch calculation takes about 1.75 times as long as one Wald calculation.

## 7.2 What the correction can and cannot change

Both methods compute the same  $\hat{\Delta}_{BC}$ ,  $\widehat{SE}$  and  $T$ . With valid positive degrees of freedom, Expanded Welch has a heavier-tailed reference and a two-sided p-value no smaller than Wald's for the same table pair. This gives a direct explanation for many plots: the correction lowers rejection whether the Wald rate is too high or already too low. It is not an adaptive calibration method that can move the threshold in both directions. It also does not alter the leading bias correction or re-estimate the numerator's finite-sample distribution.

The estimated MI difference and its standard error come from the same count tables. Independence between the two groups lets us add their variance contributions, but it does not make the MI difference independent of its estimated standard error. The diagnostic in Chapter 6 shows that both can differ from their large-sample approximations. Adjusting the reference for the standard error alone therefore cannot guarantee an exact Student distribution or correct every small-sample error.

Near independence, the calculation in Chapter 4 gives an approximate component degree of freedom of  $nI$ . When MI is small, that number can remain small even with a sizeable sample. The resulting Student reference requires a larger observed difference to reject, just where the large-sample approximation to MI is also least reliable.

At exact independence, the usual first-order MI variance is zero. An MI estimate from a finite sample can still vary, but a different, second-order approximation is then needed. Comparing two positive MI values and testing one table for independence are therefore different problems. Appendix D gives the mathematical distinction.

## 7.3 Interpretation of invalid samples

The validity diagnostics matter most where expected counts are very small. Wald needs a usable total estimated variance. The expanded implementation additionally requires each component sensitivity variance and effective degree of freedom to

be positive and finite. A sample can therefore yield a valid Wald p-value but no Expanded Welch p-value. Part of this extra invalidity comes from the implementation rule rather than the combined formula, as shown in Appendix A. The primary unconditional rate records a failed calculation as a non-rejection.

That rate answers an operational question: how often does the implemented procedure produce a valid rejection? It should not be mistaken for the conditional rejection probability among valid tables. Nor does a failure to return a p-value constitute a true negative. The saved validity and paired results allow readers to distinguish fewer rejections from fewer usable tests.

## 7.4 Practical conclusion and limits of the evidence

For the tested task, Normal Wald remains the simpler general baseline. Expanded Welch can be useful as a conservative sensitivity analysis in a regime where its validity is high and Wald's null rejection is demonstrably liberal. It should not be installed as a universal replacement merely because one null table shows a smaller false-positive rate. In applications, the population truth is unknown, so the most relevant calibration study is one matched to the anticipated alphabet, sample size and plausible margins. Selecting between the methods after inspecting a single p-value is not a validated selection rule.

Several limitations bound this conclusion. The population families are deliberately constructed fixed examples, not a probability sample of all joint distributions. Twenty thousand replicates make a rate at a fixed population relatively precise; they do not make the 534 selected population pairs representative of every application. The tables are at most  $8 \times 8$  square and  $3 \times 5$  rectangular, with the stated finite sample ladders and only a two-sided nominal level of 0.05. The main construction emphasises a local four-cell association; spread, rare-cell and log-linear checks probe but do not exhaust other possibilities. The study does not validate paired observations, continuous estimators, a real-data workflow or growing alphabets.

The protocol was frozen before the final simulation, after exploratory development of the method and design. It is a reproducible final run without external preregistration. Configurations displayed in several sections are the same sampled run. Repeating their figure appearances does not provide independent confirmation. The recorded Monte Carlo intervals also address repeated sampling of each fixed pair, not the broader uncertainty of generalising to an untested population.

The alternative curves use the common nominal 0.05 threshold. They show practical detection under the tested procedures, not power after giving each method

a simulation-specific critical value.

Future work should address the joint finite-sample behaviour of the numerator and denominator or use a calibrated reference tailored to sparse tables. The transition near independence is a separate problem because the first-order variance tends to zero. A likelihood-ratio or resampling alternative must be assessed when  $I(P) = I(Q)$  but  $P \neq Q$ , not imported from a test of full distributional equality or one-table independence without checking its target. These are directions for new validation, not remedies established by the present results.

# Chapter 8

## Conclusion

This thesis compared two tests of whether independent groups have the same population MI. Equal MI does not require independence within either group or identical population tables. Normal Wald and Expanded Welch calculate the same estimated difference and standard error; they use different reference distributions to decide whether that difference is unusual.

Expanded Welch chooses its Student reference by calculating how the estimated MI variance responds when a count table changes. The calculation includes changes to cell probabilities and to the row and column totals used in MI. It provides a reasoned way to choose degrees of freedom, but not an exact small-sample distribution.

The experiment found both benefits and costs. Expanded Welch can bring an excessive Wald false-positive rate closer to 5%. If Wald already rejects too rarely, Expanded Welch lowers rejection further. In sparse samples it also returns fewer usable results, partly because of the implementation's validity rule. A low rejection rate caused by failed calculations should not be mistaken for good calibration. Both methods approached 5% in some tested large-sample populations, but populations close to independence remained difficult. Expanded Welch took about 1.75 times as long per test in the measured implementation.

The difficulty near independence helps explain why the correction is not a general solution. It can choose a Student reference that requires stronger evidence to reject, while leaving any error in the estimated MI difference or standard error unchanged. The evidence therefore supports Normal Wald as the simpler general baseline. Expanded Welch is an additional check where Wald is known to reject equal MI too often, not a universal replacement.

# Appendix A

## Additional Derivation Details

This appendix records algebra that supports Chapters 3 and 4 but is not required to follow the main argument.

### A.1 Gradient of mutual information on the probability simplex

Mutual information can be written as

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (\text{A.1})$$

Treating the cell probabilities as coordinates gives

$$\frac{\partial I(P)}{\partial p_{ij}} = (\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \quad (\text{A.2})$$

$$= \log \frac{p_{ij}}{p_{i+} p_{+j}} - 1 \quad (\text{A.3})$$

$$= \ell_P(i, j) - 1. \quad (\text{A.4})$$

Any valid probability perturbation  $h = (h_{ij})$  satisfies  $\sum_{i,j} h_{ij} = 0$ . Its directional derivative is therefore

$$\left. \frac{d}{d\varepsilon} I(P + \varepsilon h) \right|_{\varepsilon=0} = \sum_{i,j} \{\ell_P(i, j) - 1\} h_{ij} \quad (\text{A.5})$$

$$= \sum_{i,j} \ell_P(i, j) h_{ij}. \quad (\text{A.6})$$

The constant in the unconstrained gradient disappears on the simplex. For the one-cell direction  $h_{ij} = \mathbf{1}\{i = x, j = y\} - p_{ij}$ , Equation (A.6) becomes

$$\ell_P(x, y) - \sum_{i,j} p_{ij} \ell_P(i, j) = \ell_P(x, y) - I(P), \quad (\text{A.7})$$

which is Equation (4.18).

## A.2 Why the variance of pointwise MI appears

For empirical cell probabilities,

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\}. \quad (\text{A.8})$$

Substitution into the first-order Taylor expansion of MI around  $P$  gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i, j) (\hat{p}_{ij} - p_{ij}) \quad (\text{A.9})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \left[ \sum_{i,j} \ell_P(i, j) \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\} \right] \quad (\text{A.10})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \{\ell_P(Z_a^{(P)}) - I(P)\}. \quad (\text{A.11})$$

The observation-level summand has variance

$$\text{Var}_P\{\ell_P(X, Y) - I(P)\} = \mathbb{E}_P \left[ \{\ell_P(X, Y) - I(P)\}^2 \right] \quad (\text{A.12})$$

$$= \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2 \quad (\text{A.13})$$

$$= V(P). \quad (\text{A.14})$$

This is why the first-order variance of the plug-in MI estimator is  $V(P)/n_P$ , not a variance of the numerical category labels.

## A.3 Mean of the variance influence function

The influence function  $g_P(x, y)$  is centred at the population level. This can be shown without expanding its individual terms. The probability-weighted average of

the one-cell perturbation directions is

$$\sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\} = P - P = 0. \quad (\text{A.15})$$

The first derivative is linear in the direction of change. Therefore, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$E_P\{g_P(X, Y)\} = \sum_{x,y} p_{xy} \left. \frac{d}{d\varepsilon} V\{(1 - \varepsilon)P + \varepsilon\delta_{(x,y)}\} \right|_0 \quad (\text{A.16})$$

$$= \left. \frac{d}{d\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\}\right\} \right|_0 \quad (\text{A.17})$$

$$= \left. \frac{d}{d\varepsilon} V(P) \right|_0 = 0. \quad (\text{A.18})$$

The same weighted cancellation holds after exact plug-in substitution. The implementation nevertheless subtracts the probability-weighted empirical mean inside Equation (4.38) to protect the variance calculation from floating-point roundoff; this does not change the population definition.

## A.4 Direct table substitutions

For observed counts  $N^P = (N_{ij}^P)$ , the quantities used by expanded Welch are calculated as

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{p}_{i+} = \sum_j \hat{p}_{ij}, \quad \hat{p}_{+j} = \sum_i \hat{p}_{ij}, \quad (\text{A.19})$$

$$\hat{\ell}_P(i, j) = \log \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}}, \quad \hat{I}(P) = \sum_{i,j} \hat{p}_{ij} \hat{\ell}_P(i, j), \quad (\text{A.20})$$

$$\hat{V}(P) = \sum_{i,j} \hat{p}_{ij} \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2. \quad (\text{A.21})$$

All sums involving pointwise MI are restricted to cells with  $\hat{p}_{ij} > 0$ . Direct substitution gives

$$\hat{g}_P(i, j) = \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \quad (\text{A.22})$$

$$+ 2 \left\{ \hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right\}, \quad (\text{A.23})$$

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \left\{ \hat{g}_P(i, j) - \sum_{k,l} \hat{p}_{kl} \hat{g}_P(k, l) \right\}^2, \quad (\text{A.24})$$

$$\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}. \quad (\text{A.25})$$

These expressions are the direct  $O(rc)$  implementation of the population derivation.

## A.5 Dependence between the MI and variance estimates

Both  $\hat{I}(P)$  and  $\hat{V}(P)$  are calculated from the same sample. Under fixed positive support, their first-order covariance is

$$\text{Cov}\{\hat{I}(P), \hat{V}(P)\} \approx \frac{1}{n_P} \mathbb{E}_P[\{\ell_P(X, Y) - I(P)\}g_P(X, Y)]. \quad (\text{A.26})$$

The corresponding expression for  $Q$  replaces  $P$  and  $n_P$  with  $Q$  and  $n_Q$ . This covariance need not be zero. Matching the variability of the estimated variance alone therefore does not make the test statistic exactly Student distributed.

## A.6 Reference ordering and zero components

For positive component degrees of freedom, substituting Equation (4.39) and its  $Q$  analogue into Equation (4.40) gives the equivalent combined form

$$\hat{\nu}_{\text{expanded}} = \frac{\{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q\}^2}{\hat{\tau}^2(P)/(2n_P^3) + \hat{\tau}^2(Q)/(2n_Q^3)}. \quad (\text{A.27})$$

This direct form remains defined if one table has  $\hat{V} = \hat{\tau}^2 = 0$  and the other supplies a positive denominator contribution. The zero table drops out, leaving the other table's component degree of freedom. The evaluated implementation instead requires both component degrees of freedom to be positive and finite before combining them. It records such a one-zero-component case as invalid, even though the direct form has a value.

The Student p-value is no smaller than the Normal Wald p-value for the same finite statistic. To see this, write a Student variable as  $Z/\sqrt{S}$ , where  $Z \sim N(0, 1)$ ,  $S \sim \chi_\nu^2/\nu$ , and  $Z$  and  $S$  are independent. For  $c > 0$ , the conditional two-sided tail  $2\{1 - \Phi(c\sqrt{s})\}$  is strictly convex in  $s$ , since its second derivative is  $c\phi(c\sqrt{s})(s^{-3/2} + c^2s^{-1/2})/2 > 0$ . As  $\mathbb{E} S = 1$ , Jensen's inequality gives  $\Pr(|Z/\sqrt{S}| > c) > \Pr(|Z| > c)$ ; at  $c = 0$  both tails equal one. Moreover, the implemented Expanded-valid set is a subset of the Wald-valid set. At a common fixed threshold, every Expanded rejection

is therefore also a Wald rejection, both replicate by replicate and in unconditional rates that count invalid outputs as non-rejections.

## A.7 Numerical derivative checks

The implementation tests the analytic derivatives against centred finite differences on positive probability tables. For a selected cell and a small positive step  $\eta$ , define

$$P_{\pm\eta} = (1 \mp \eta)P \pm \eta\delta_{(x,y)}. \quad (\text{A.28})$$

For sufficiently small  $\eta$ , the checks compare

$$\frac{I(P_\eta) - I(P_{-\eta})}{2\eta} \quad \text{with} \quad \ell_P(x, y) - I(P), \quad (\text{A.29})$$

$$\frac{V(P_\eta) - V(P_{-\eta})}{2\eta} \quad \text{with} \quad g_P(x, y). \quad (\text{A.30})$$

The finite-difference step is selected so that both perturbed tables retain positive probabilities. These checks test the algebra independently of the Monte Carlo calibration experiment.

# Appendix B

## Population Construction and Exact Design Settings

The experiment uses fixed population tables. This appendix states the construction precisely enough to reproduce any plotted point and records the finite parameter grids. Full-precision probabilities for every constructed pair are stored in `population_definitions.csv`; rounded examples in the thesis are for reading, not simulation input.

### B.1 Additive construction

For prescribed positive row vector  $a$  and column vector  $b$ , form  $B = ab^\top$ . The principal first-cell direction  $H$  contains  $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$  in rows and columns 1 and 2 and zero elsewhere. The rare-cell direction places the same block in the last two rows and columns. The spread direction is

$$H_{ij} = u_i v_j, \quad u_i = -1 + \frac{2(i-1)}{r-1}, \quad v_j = -1 + \frac{2(j-1)}{c-1}. \quad (\text{B.1})$$

Both score vectors sum to zero, so every direction preserves margins.

For  $t \geq 0$ , define  $R(t) = B + tH$ . Calculate the first strength at which a decreasing cell reaches zero,

$$t_{\max} = \min_{H_{ij} < 0} \frac{B_{ij}}{-H_{ij}}. \quad (\text{B.2})$$

Evaluate the path endpoint MI. A target MI must lie strictly below this endpoint, within the stated numerical tolerance. For a positive target, Brent's one-dimensional root search ([Brent, 1973](#)) solves  $I\{R(t)\} = I_{\text{target}}$  on  $[0, t_{\max}]$ ; a zero target uses

$t = 0$ . The final table is accepted only if every cell is strictly positive, the sum is one, both margins agree with their prescribed vectors, and achieved MI agrees with the target. The normalisation and margin tolerances are  $10^{-12}$ ; the MI tolerance is  $10^{-12} + 10^{-8}I_{\text{target}}$ . These numerical checks ensure that a horizontal-axis difference denotes constructed population truth, not an empirical estimate. They do not impose an expected-count floor on the subsequent sample.

The row and column vectors are separately constructed for each population. For shape  $r \times c$ , profile **uniform** uses uniform vectors of lengths  $r$  and  $c$  for both  $P$  and  $Q$ . Profile **same\_skew** uses  $m_r(0.8)$ ,  $m_c(0.8)$  for both. Profile **different\_skew** uses  $m_r(0.7)$ ,  $m_c(0.7)$  for  $P$  and  $m_r(0.8)$ ,  $m_c(0.8)$  for  $Q$ . The extreme regimes instead state the two dominant probabilities directly. The direction of the alternative is a separate parameter and should not be inferred from which population has the more skewed margin.

The log-linear comparison uses the scores  $u_i, v_j$  from Equation (B.1), but starts from positive cell weights

$$K_{ij}(\lambda) = \exp(\lambda u_i v_j) \quad (\text{B.3})$$

instead of adding  $tH$  to the independence table  $B$ . Iterative proportional fitting (Deming and Stephan, 1940) scales rows and columns of these weights until the prescribed margins are restored. At  $\lambda = 0$ , this gives the independence table; a numerical search then chooses  $\lambda$  to reach the target MI. The final table passes the same positivity, normalisation, margin and target-MI checks as the additive table. Unlike the first-block additive construction used in the comparison, it changes cells throughout the table. Matching margins and MI does not imply matching all cell probabilities or expected counts.

## B.2 Frozen statistical grids

Unless a row of Table B.1 says otherwise, both populations use the first-cell additive direction,  $I(P) = I_0$ ,  $I(Q) = I_0 + |\Delta|$ , and the three standard margin profiles. The symbol  $n$  means  $n_P = n_Q = n$ . A requested point was simulated only after the population constructor verified a strictly positive feasible table. The final display contains no infeasible slots, but that is a result of choosing feasible finite grids. Curves should not be extrapolated beyond their last constructed target.

**Table B.1:** Exact statistical experiment grids; MI values are in nats.

| Family           | Settings                                                                                                                                                                                                                                                                        |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main             | Shapes $2 \times 2, 3 \times 3, 5 \times 5, 8 \times 8$ ; three profiles; $I_0 = 0.02$ ; $ \Delta  \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02\}$ ; $n \in \{2, 5, 10, 20, 50, 100, 250, 500, 1000\}$ .                                                                            |
| Baseline         | Main shapes and profiles; $I_0 \in \{0.0001, 0.001, 0.02\}$ ; main difference grid; $n \in \{20, 100, 1000\}$ .                                                                                                                                                                 |
| Changed cells    | Shapes $3 \times 3, 5 \times 5, 8 \times 8$ ; different-skew margins; $I_0 = 0.0001$ ; $ \Delta  \in \{0, 0.0001, 0.00025, 0.0005, 0.001, 0.002\}$ ; $n \in \{20, 100, 1000\}$ . First and rare blocks for all shapes; spread pattern for $5 \times 5$ and $8 \times 8$ .       |
| Imbalance        | Main shapes, profiles and difference grid; $I_0 = 0.02$ . Sample pairs are $(50, 50), (50, 100), (100, 50), (50, 250), (250, 50), (50, 500), (500, 50), (500, 500)$ . Both $Q$ -higher and $P$ -higher MI directions.                                                           |
| Larger effects   | Shapes $2 \times 2, 3 \times 3$ ; three profiles; $I_0 = 0.02$ ; main difference grid plus $0.05, 0.1, 0.2$ ; $n \in \{20, 100, 1000\}$ .                                                                                                                                       |
| Rectangular      | Shapes $2 \times 3, 3 \times 5$ ; three profiles; $I_0 = 0.02$ ; main difference grid; $n \in \{20, 100, 1000\}$ .                                                                                                                                                              |
| Construction     | Shapes $3 \times 3, 5 \times 5, 8 \times 8$ ; uniform and different-skew profiles; $I_0 \in \{0.0001, 0.02\}$ ; main difference grid; $n \in \{20, 100, 1000\}$ ; additive first block and ordinal log-linear constructors.                                                     |
| Extreme skew     | Shapes $2 \times 2, 3 \times 3, 8 \times 8$ ; dominant margin probability pairs $(\rho_P, \rho_Q) \in \{(0.9, 0.95), (0.99, 0.995), (0.999, 0.9995)\}$ ; $I_0 = 0.00001$ ; $ \Delta  \in \{0, 0.00001, 0.00005, 0.0001, 0.0002\}$ ; $n \in \{1, 2, 5, 20, 100, 1000, 10000\}$ . |
| Rare-cell stress | Shapes $3 \times 3, 8 \times 8$ ; dominant pair $(0.9, 0.95)$ ; first and rare blocks; $I_0 = 0.00001$ ; $ \Delta  \in \{0, 0.00001, 0.000025, 0.00005, 0.0001\}$ ; same $n$ ladder as extreme skew.                                                                            |
| Independence     | Shapes $2 \times 2, 8 \times 8$ ; three profiles; $I_0 = 0$ ; main difference grid; $n \in \{2, 20, 100, 1000\}$ .                                                                                                                                                              |

---

| Family      | Settings                                                                                                                                                                                                                                     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Convergence | Main square shapes and profiles; equal MI $I(P) = I(Q) \in \{0.0001, 0.02\}$ ; $n \in \{1000, 2500, 10000, 50000\}$ . Additional rare-block controls use $3 \times 3, 5 \times 5, 8 \times 8$ , different-skew margins and common MI 0.0001. |

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The  $n = 1$  stress setting is retained and not hidden; the statistic is not evaluable there and its result is recorded as invalid. Every unique statistical configuration has 20,000 sampled table pairs. When the same population pair and sample sizes appear in another section, the saved configuration is reused for presentation and not simulated again.

# Appendix C

## Evidence Index and Additional Diagnostics

The main results chapter shows a systematic selection of full-scale and calibration-zoom figures. The full companion atlas remains the source for every sample size and focused regime in the frozen design. The index below specifies how to retrieve exact numbers without averaging regimes into a single summary. Paths are relative to the workspace root `Simulations/`. The final results directory used below is `WelchSatterthwaiteMI/results/thesis_redesign/`. These are workspace paths; Appendix E gives commands for the extracted archive, where `main.tex` and `WelchSatterthwaiteMI/` share one root.

The companion written atlas is at `WelchSatterthwaiteMI/docs/experiments/THESIS_EXPERIMENTS.md`. Its 99 primary figures and seven extra calibration zooms retain the full design. The thesis-specific PDFs in `Thesis Writeup/Welch MI/figures_rewrite/` are generated from the two final CSV files named in the source script. The adjacent `figure_manifest.json` lists, for each thesis PDF, its original section, source files, configuration identifiers and 20,000-replicate denominator. The file `evidence_values.tex` is generated from exact configuration IDs for the numerical examples in Chapter 6.

For each configuration, 20,000 table pairs are sampled. The plotted rate is the number of valid rejections divided by 20,000. The method-valid conditional rate divides valid rejections by that method's valid outputs; it is undefined if there are none. The common-valid rate restricts to outputs where both methods are valid. A paired difference is based on counts of the four joint rejection outcomes for the same table pairs. These distinctions allow a reader to check whether a curve difference is primarily a changed reference or lost validity.

For the paired comparison, let  $D_s$  equal the Wald valid-rejection indicator minus

**Table C.1:** Saved files and the question each answers.

| File in the final results directory                                      | Use                                                                                                                                                                                |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>display_manifest.csv</code>                                        | Figure section, row, column, horizontal value and exact configuration identifier; a configuration may recur.                                                                       |
| <code>configuration_manifest.csv</code>                                  | Unique statistical configurations, sample sizes, target populations and simulation identifiers.                                                                                    |
| <code>population_definitions.csv</code>                                  | Full-precision $P$ and $Q$ probability tables, margins, achieved MI, constructor and feasibility diagnostics for every pair.                                                       |
| <code>cell_results.csv</code>                                            | One row per configuration and method: unconditional rejection, method validity, conditional and common-valid rates, Wilson intervals, expected-count and support-loss diagnostics. |
| <code>paired_method_results.csv</code>                                   | Paired rejection counts, rate differences and approximate paired intervals from the same replicates.                                                                               |
| <code>runtime_summary.csv</code>                                         | Per-regime median and interquartile call times in microseconds, validity and Expanded-to-Wald time ratio.                                                                          |
| <code>verification.json</code> and <code>report_verification.json</code> | Run-level population, denominator, pairing, method and report checks.                                                                                                              |

the Expanded Welch valid-rejection indicator for sampled table pair  $s$ , matching the sign used in Table 6.1. Its average is  $\bar{D} = (1/20,000) \sum_{s=1}^{20,000} D_s$ , with approximate 95% interval

$$\bar{D} \pm 1.96 \left\{ \frac{1}{20,000(19,999)} \sum_{s=1}^{20,000} (D_s - \bar{D})^2 \right\}^{1/2}. \quad (\text{C.1})$$

This calculation preserves the within-replicate pairing. In the saved CSV, the generic field is stored as method A minus method B, with Expanded Welch listed as method A; Table 6.1 therefore reports its negative. The individual method intervals use Wilson’s binomial score construction (Wilson, 1927).

For any  $P$  population with sample size  $n_P$ , its minimum expected cell count is  $\min_{ij} n_P p_{ij}$ ; define  $Q$ ’s analogously. No “expected minimum” of random observed counts is calculated. The saved support diagnostics include empty observed rows and columns and the fraction of zero-count cells. Such values describe why a statistic might fail, but do not constitute a universal validity threshold.

The thesis does not treat a descriptive count of regimes closer to 0.05 as a formal paired test over regimes. Wilson intervals quantify binomial Monte Carlo variation at each fixed point, and paired normal intervals quantify a specified within-configuration method difference. Neither interval family is a simultaneous assertion across all configurations. Tables sharing a population but changing  $n$  or  $|\Delta|$  answer different fixed-design questions and should not be collapsed into one mean rate.

# Appendix D

## The Independence Boundary

The main thesis compares populations with  $I(P) = I(Q) > 0$ , allowing  $P \neq Q$ . This appendix explains why constructing an independent reference distribution does not turn the Expanded Welch method into a new first-order test of independence.

### D.1 Constructing the independent reference

For a joint distribution  $P = (p_{ij})$ , define

$$q_{ij} = p_{i+}p_{+j}. \quad (\text{D.1})$$

Then  $Q = (q_{ij})$  is a probability distribution because

$$\sum_{i,j} q_{ij} = \left( \sum_i p_{i+} \right) \left( \sum_j p_{+j} \right) = 1. \quad (\text{D.2})$$

It has the same margins as  $P$ , since

$$\sum_j q_{ij} = p_{i+}, \quad \sum_i q_{ij} = p_{+j}. \quad (\text{D.3})$$

The factorisation in Equation (D.1) makes  $X$  and  $Y$  independent under  $Q$ , so  $I(Q) = 0$ .

The divergence from  $P$  to this reference is

$$D_{\text{KL}}(P \| Q) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (\text{D.4})$$

$$= \sum_{i,j} p_{ij} \log \frac{p_{ij}}{p_{i+}p_{+j}} \quad (\text{D.5})$$

$$= I(P). \quad (\text{D.6})$$

Consequently, testing  $I(P) = I(Q)$  for this constructed reference is exactly testing  $I(P) = 0$ , or independence.

## D.2 Why the first-order term vanishes

At independence,  $p_{ij} = p_{i+}p_{+j}$ , and every population pointwise-MI value is zero:

$$\ell_P(i, j) = \log 1 = 0. \quad (\text{D.7})$$

Equation (4.18) therefore gives

$$\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P) = 0. \quad (\text{D.8})$$

The first-order variance also vanishes:

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\} = 0. \quad (\text{D.9})$$

The denominator used by Normal Wald and Expanded Welch is thus degenerate under the population independence null. This does not mean that a finite sample must have zero plug-in MI or zero plug-in variance. Sampling noise usually makes the empirical table depart from exact independence. The point is that these random departures do not enter MI through the usual first-order term at the null. They enter through the second-order term derived next, so changing only the degrees of freedom of a first-order statistic cannot supply the missing source of variation.

## D.3 Second-order behaviour

Fix a strictly positive independent table  $P$ , and let  $h = (h_{ij})$  describe a small redistribution of probability, with  $\sum_{i,j} h_{ij} = 0$ . Define the path

$$P_\varepsilon = P + \varepsilon h, \quad f(\varepsilon) = I(P_\varepsilon), \quad (\text{D.10})$$

for values of  $\varepsilon$  small enough that all cells remain positive. Here  $\varepsilon = 0$  is the expansion point and  $f(0) = I(P) = 0$ . The second-order Taylor expansion is

$$f(\varepsilon) \approx \underbrace{f(0)}_{\text{zeroth order}} + \underbrace{f'(0)\varepsilon}_{\text{first order}} + \underbrace{\frac{f''(0)}{2!}\varepsilon^2}_{\text{second order}}. \quad (\text{D.11})$$

The first-order term is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = 0. \quad (\text{D.12})$$

Both parts vanish:  $\ell_P(i,j) = 0$  at independence, and the probability constraint gives  $\sum_{i,j} h_{ij} = 0$ . Thus, positive and negative cell errors cannot create a linear change in MI around the independence table.

To state the surviving term, define the row and column changes  $h_{i+} = \sum_j h_{ij}$  and  $h_{+j} = \sum_i h_{ij}$ . Differentiating the entropy form of MI twice gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \quad (\text{D.13})$$

Because  $p_{ij} = p_{i+}p_{+j}$ , define the part of each cell change that cannot be explained by changed margins as

$$a_{ij} = h_{ij} - p_{+j}h_{i+} - p_{i+}h_{+j}. \quad (\text{D.14})$$

Equation (D.13) can then be written as

$$f''(0) = \sum_{i,j} \frac{a_{ij}^2}{p_{i+}p_{+j}}. \quad (\text{D.15})$$

Substituting the zeroth-, first-, and second-order terms into Equation (D.11) gives

$$I(P_\varepsilon) = \underbrace{0}_{\text{zeroth and first order}} + \underbrace{\frac{\varepsilon^2}{2} \sum_{i,j} \frac{a_{ij}^2}{p_{i+}p_{+j}}}_{\text{second order}} + o(\varepsilon^2). \quad (\text{D.16})$$

The remainder  $o(\varepsilon^2)$  is negligible relative to  $\varepsilon^2$  as  $\varepsilon$  approaches zero.

For a multinomial sample, take  $\varepsilon h = \hat{P} - P$ . An empirical cell proportion has standard deviation  $\sqrt{p_{ij}(1-p_{ij})/n}$ , so each  $\varepsilon h_{ij}$  is typically proportional to  $n^{-1/2}$ . The corresponding row- and column-adjusted errors  $\varepsilon a_{ij}$  have the same scale. Equation (D.16) squares these errors, making  $\hat{I}(P)$  proportional to  $n^{-1}$  under independence. Multiplying by  $2n$  therefore produces a statistic with a non-vanishing scale:

$$2n\hat{I}(P) \xrightarrow{d} \chi_{(r-1)(c-1)}^2. \quad (\text{D.17})$$

This is the usual likelihood-ratio, or  $G$ , test of independence under a fixed alphabet and the standard positive-support large-sample conditions (Wilks, 1938; Brillinger, 2004). The chi-squared reference describes the squared cell errors in Equation (D.16).

Expanded Welch instead standardises a signed first-order MI error on the  $n^{-1/2}$  scale. At independence that signed term is absent, so its Student reference is not the chi-squared reference in another form.

Although squaring a regular asymptotic  $N(0, 1)$  statistic gives a  $\chi_1^2$  limit, the present  $T$  is degenerate at independence. Its square is therefore not the likelihood-ratio  $G$  statistic.

The same likelihood-ratio connection appears for conditional information measures. For finite Markov chains, [Barnett and Bossomaier \(2012\)](#) relate transfer entropy to a log-likelihood ratio with an asymptotic chi-squared reference; the wider information-theoretic setting is reviewed by [Bossomaier et al. \(2016\)](#). This is an adjacent application of the same second-order likelihood principle, not a two-sample equal-MI Welch test.

## D.4 What alternative constructions produce

Using the observed margins to set  $\hat{q}_{ij} = \hat{p}_{i+}\hat{p}_{+j}$  gives

$$2nD_{\text{KL}}(\hat{P}||\hat{Q}) = 2n\hat{I}(P), \quad (\text{D.18})$$

which is the likelihood-ratio statistic above. Because  $\hat{Q}$  is constructed from the same sample as  $\hat{P}$ , it is not an independent second table and cannot be inserted into the two-sample Welch variance formula as though it supplied new information.

Repeatedly shuffling one variable preserves its observations while breaking the pairing with the other variable. The resulting empirical distribution is a permutation reference. Repeatedly drawing tables from  $\hat{P}_X\hat{P}_Y$  is a parametric bootstrap reference. A single simulated  $Q$  table adds Monte Carlo variation but does not by itself define a calibrated null distribution.

The independent reference  $Q = P_X P_Y$  is therefore mathematically valid, but it does not create a regular two-sample MI comparison. Analytic improvement of the sparse independence test would require a second-order finite-sample method and is not an immediate extension of the Expanded Welch calculation.

# Appendix E

## Reproducibility and Protocol Record

The final protocol is `WelchSatterthwaiteMI/experiments/THESIS_REDESIGN_PROTOCOL.json`, version 1.0.0, frozen on 7 September 2026 UTC before the final simulation. It specifies the population grids, seeds, methods, 20,000 replicates,  $\alpha = 0.05$ , validation tolerances and timing design. This local freeze followed exploratory development; it was not an external preregistration.

The run contains 3,111 unique configurations, 4,001 display positions and 534 population pairs, corresponding to 62,220,000 sampled table pairs. The saved verification records check the population targets, result denominators, paired method outputs, validity rules and generated atlas. Chapter 6 uses the same frozen tables; no population was resimulated for a thesis figure.

The software requires matching table shapes, at least two rows and columns, finite nonnegative integer counts, and more than one observation in each table. Wald requires a finite corrected difference and standard error,  $\hat{V}(P) + \hat{V}(Q) > 10^{-14}$ , and a finite p-value. Expanded Welch also requires positive finite component and combined degrees of freedom and a finite p-value. It does not calculate a combined degree of freedom when a component degenerates, even if the direct combined expression in Appendix A has a value. Failed conditions are recorded as invalid calculations.

The exact software versions, seeds and source hashes are recorded in `WelchSatterthwaiteMI/results/thesis_redesign/run_metadata.json`. The experiment supplement contains the protocol, implementation, consolidated results, 3,111 configuration checkpoints, complete atlas, mechanism diagnostics and tests. Its README documents minor post-run orchestration changes; the frozen population construction, simulation and method calculations are unchanged.

The evidence audit reconstructs the protocol manifests and checks every figure and Chapter 6 claim against the saved files. Final checksums are in `SHA256SUMS.txt`. After extracting the source and supplement into one archive root, run these commands from the directory containing `main.tex` and `WelchSatterthwaiteMI/`:

```
python figures_rewrite/make_figures.py
python figures_rewrite/audit_evidence.py
latexmk -pdf main.tex
```

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